

6

Limit Order Book Markets

Learning Objectives:

- Why limit order markets require separate analysis
- Adverse selection and market depth in limit order markets
- Effects of trading rules on market liquidity in limit order markets
- Determinants of the choice between market and limit orders

6.1 Introduction

In many markets, trading occurs in continuous time, via an electronic limit order book (LOB), as Chapter 2 describes. For instance, nearly all the major equity markets (the NYSE, Nasdaq, Euronext, the London Stock Exchange, Deutsche Börse, etc.) are continuous limit order markets, sometimes in combination with other trading mechanisms such as a dealer market. Many other instruments (bonds, currencies, derivatives) are also traded in continuous limit order markets. For instance, LIFFE (London International Financial Futures and options Exchange) and the CME (Chicago Mercantile Exchange) use this mechanism to trade derivatives. Likewise, MTS (a trading platform for European sovereign bonds) and BrokerTec (an interdealer trading platform for US Treasuries) are organized as continuous limit order markets.

Limit order markets deserve specific analysis for a number of reasons. First, their trading rules differ from those in call markets or dealership markets, analyzed in Chapter 4. The limit order market is a *discriminatory auction*: all limit orders filled in a given transaction are executed at their own posted price. In contrast, the call market is a *uniform auction*: all traders involved in a given transaction receive or pay the same price.

The following example (Figure 6.1) illustrates the difference. One hundred shares are offered for sale at $A_1 = \$100.50$, two hundred additional shares are offered at $A_2 = \$101$, and an unlimited number of shares is offered for sale at $A_3 = \$101.50$. A buy market order for three hundred shares arrives. It triggers the execution of the sell limit orders placed at \$100.50 and \$101. In the call

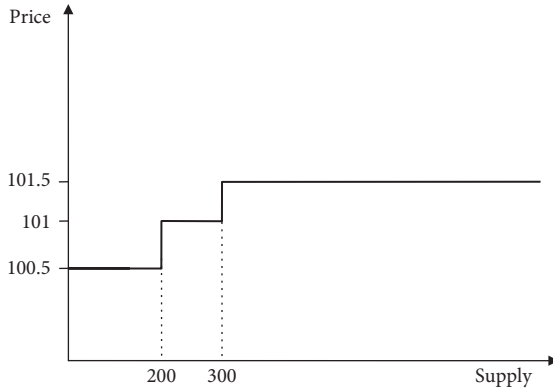


Figure 6.1 A limit order book.

market, all sell limit orders eligible for execution execute at \$101, the marginal execution price for the market order. The buyer's payment is therefore \$30,300. But in the continuous limit order market, the buy market order "walks up" the book and each limit order executes at its own price. That is, one hundred shares are bought at \$100.5 and two hundred shares at \$101. The buyer then pays \$30,250.

Clearly, for a given LOB, traders submitting market orders get a better deal in the discriminatory auction. However, participants will typically use different order placement strategies in the two types of market since execution prices are determined differently. Chapter 4 analyzes the behavior of investors in a call market. Here, we examine their behavior in a continuous limit order market. To this end, Section 6.2 analyzes a static model of the optimal bidding strategies for limit order traders, maintaining the assumption that traders submitting market and limit orders are different. Thus, this model extends the canonical models considered in Chapters 3 and 4, accounting for the fact that limit order markets operate differently from call auctions. Section 6.3 applies the model to the design of limit order markets: it analyzes the impact of tick size, the role of priority rules and that of designated liquidity suppliers. This model will also be used in Chapter 7 to study how competition between trading platforms affects liquidity (Section 7.4.2).

Another reason why limit order markets deserve a specific analysis is that they feature no strict dichotomy between the traders who "make" the market by posting quotes and those who "take" these quotes by hitting them: any participant can carry out the desired trades by submitting limit orders, market orders, or a combination. Thus, the provision of liquidity does not rest exclusively with

a designated group of market makers (as assumed in the models described in Chapters 3 and 4): in these markets, liquidity depends on how agents trade off the costs and benefits of limit orders and market orders.

The benefit of limit orders is straightforward: they enable investors to earn revenues from liquidity provision rather than pay a cost for liquidity consumption. However, limit orders are costly for two reasons. First, limit orders' users are exposed to the risk of losing money to better informed agents, very much like market makers (see Chapter 3). This happens in particular when news arrives to the market. In this case, if limit order traders do not update (or cancel) their limit orders fast enough, they face the risk of trading at a stale price with investors who react to the news faster. This risk, which is called the risk of being picked off, has become particularly acute with the emergence of high-frequency trading (see Chapter 9). Second, limit order users face the risk of non-execution. For instance, a sell limit order is less likely to be filled if its price is not low enough. Non-execution is costly because order management takes time (sometimes it even requires the payment of fees to enter and modify orders), and because it can have opportunity costs (see the discussion about the implementation shortfall in Chapter 2). In Section 6.4 we analyze how these costs affect the pricing of limit orders (hence, the liquidity available to traders who place them) and the choice between market and limit orders. Understanding this choice is important, because the viability of limit order markets depends critically on traders placing both types of order. In making this choice, investors trade off price improvement against the risks of non-execution and being picked off. Thus the determinants of these risks—such as the volatility of asset fundamentals—ultimately shape investors' choice between market and limit orders.

The models presented in this chapter are useful in analyzing data on trades and quotes. The data are basically of two types: snapshots of LOBs at various points in time and flows of market and limit orders over a period of time. Snapshots of the LOB enable measuring the bid-ask spread and the cumulative number of shares offered up to a given ask or bid price. That is, they describe the state of the LOB at a given point in time. The framework presented in Section 6.2 helps to explain why this state may vary from security to security or over time. Order flow data provide information on the terms of completed trades and give insights into the drivers of trading aggressiveness, market orders being more aggressive than limit orders, and limit orders' aggressiveness depending on their price. An important driver, of course, is the state of the book, which is itself influenced by the order flow. Dynamic models in which traders choose between market and limit orders offer insight into the interactions between the flow of limit and market orders and the state of the LOB (Section 6.4).

6.2 A Model of the Limit Order Book (LOB)

6.2.1 The Market Environment

We consider a limit order market for a risky security with final value $v = \mu + \varepsilon$, where $E(\varepsilon) = 0$. In period 0, traders submit limit orders. Then, in period 1, a market order arrives and executes against the limit orders in the book. In period 2, the value of the security and traders' payoffs are realized. The time line is given in Figure 6.2.

The limit orders are positioned on a price grid. The distance between two consecutive prices on the grid, the tick size, is denoted by Δ . In Figure 6.3, we denote by A_k the k^{th} price on this grid above the expected value μ of the security, and by B_k the k^{th} price on this grid below μ .

In this model, traders never submit sell limit orders at a price lower or buy limit orders at a price higher than μ , as such orders would entail expected losses if executed. For this reason, we refer to $\{B_k\}_{k=1}^{\infty}$ as the set of bid prices and to $\{A_k\}_{k=1}^{\infty}$ as the set of ask prices. Let y_k be the number of shares offered at price A_k and Y_k be the cumulative depth at price A_k (i.e., the total number of shares offered at price A_k or less). Figure 6.4 represents the ask side of the LOB.

In period 1, a trader submits a market order of size q . A buy market order executes against sell limit orders until it is fully served. Symmetrically, a sell market order executes against buy limit orders. As usual, the size of the market order is signed according to its direction: $q < 0$ for sell and $q > 0$ for buy. Market

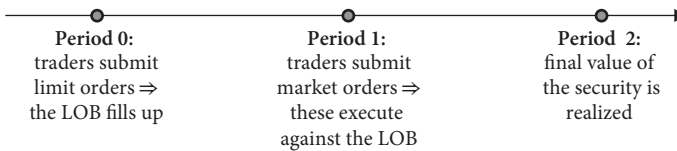


Figure 6.2 Timing.

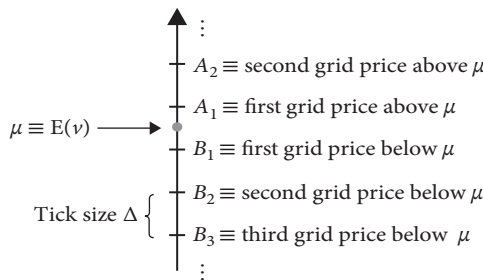


Figure 6.3 The grid of possible quotes in the limit order book.

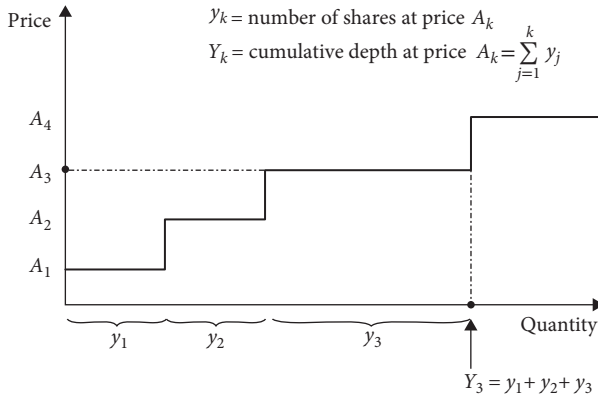


Figure 6.4 The ask side of the limit order book.

orders execute against the book at the limit order prices. Suppose a trader submits a buy market order for q shares such that: $Y_1 < q < Y_2$. His total payment is $Y_1 \cdot A_1 + (q - Y_1) \cdot A_2$.

The market order submitted in period 1 can be a buy or a sell with equal probability. Its size is unknown to the limit order traders when they submit their orders in period 0. The cumulative probability distribution of the order size is denoted by $F(\cdot)$ and its density by $f(\cdot)$. In some of the examples below, the size of market orders is assumed to be exponentially distributed, that is,

$$f(q) = \frac{1}{2} \theta e^{-\theta |q|}. \quad (6.1)$$

In this case, buy and sell market orders arrive with equal probability, since $\Pr(q > 0) = \Pr(q < 0) = \frac{1}{2}$ and the expected size of a market order, $E(|q|)$, is $\frac{1}{\theta}$.

At time 0, limit orders are submitted by a continuum of risk-neutral traders who arrive sequentially. They fill the book up to the point where there are no expected profit opportunities: a competitive equilibrium is reached when (i) there is no price at which adding a limit order is profitable (the “no entry” condition) and (ii) there is no price at which cancelling a limit order is profitable (the “no exit” condition).

Last, we assume a time-priority rule for tie-breaking.¹ This means that two limit orders at the same price are executed in the order in which they have been submitted. For instance, the last share offered at price A_k (the marginal share at this price) executes if and only if the next market order is a buy order that

¹ Other tie-breaking rules are considered in Section 6.3.2.

exceeds the number of shares offered up to price A_k , that is $q > Y_k$. Hence, the execution probability of the marginal share offered at price A_k is:

$$P(Y_k) \equiv \Pr(q \geq Y_k) = 1 - F(Y_k). \quad (6.2)$$

This execution probability declines with cumulative depth at price A_k . As the total number of shares offered up to a given price increases, it becomes less and less likely that the next market order will be large enough to trigger execution of the marginal share. This is akin to a queuing situation in which, as the queue lengthens, the last person has a smaller chance of being served because the object for sale is in unknown but limited supply. The execution probability for the marginal buy limit order at price B_k is obtained in a similar way and is equal to $F(Y_k)$, where $|Y_k|$ is the cumulative quantity demanded at bid price B_k .

This provides a simple framework for analyzing many interesting issues relating to limit order markets. However, two inherent limitations should be noted at the outset.

First, the model says nothing about how the competitive equilibrium is reached. The underlying idea is that the market is sufficiently competitive so that there are no strictly profitable opportunities for adding limit orders to the book. This is an interesting benchmark, but in the real world competition among limit order traders is likely to be imperfect. The competitive equilibrium should therefore be viewed as the limit of the equilibrium obtained when the number of market participants is infinite.

Second, in this model, traders are classed in either one of two groups: (i) those who submit limit orders (supply liquidity) and (ii) those who submit market orders (demand liquidity). They cannot switch from one group to the other. We defer the analysis of how traders choose between a market order or a limit order to Section 6.4.

6.2.2 Execution Probability and Order Submission Cost

To illustrate the notion of competitive equilibrium, we first consider a simple case in which no trader has private information about the final value of the security. Market orders are exogenous, and competing traders place limit orders at a display cost of C per share. This cost represents the time required to submit and monitor the order ("order management cost"), and any entry fee charged to traders submitting limit orders, whether or not they are filled.

This specification of the model illustrates the importance of execution probabilities for limit orders and shows how order submission costs affect cumulative depth. The mechanics here also prepare the ground for the more complex case

in which market order traders can be informed (Section 6.2.3). We consider the determination of cumulative depth on the ask side only, leaving the symmetric case of the bid side to the reader.

The zero-profit condition. We write the expected profit on the “marginal” unit offered at price A_k when cumulative depth at this price is Y_k . The trader offering this unit is called the marginal trader at price A_k .

In case of execution, the realized profit on the marginal unit is $A_k - v - C$. In case of non-execution, it is $-C$: the submission cost is paid but the order does not execute. Consequently, the expected profit, $\Pi_k(Y_k)$, on the marginal unit is:

$$\Pi_k(Y_k) = P(Y_k) [A_k - E(v|q \geq Y_k)] - C. \quad (6.3)$$

Since we assume that market order traders have no private information, the order flow at time 1 is independent of the value of the security. This means that $E(v|q \geq Y_k) = E(v) = \mu$. Hence,

$$\Pi_k(Y_k) = P(Y_k)(A_k - \mu) - C. \quad (6.4)$$

The probability of execution for a sell limit order cannot be greater than one-half, as a buy market order is submitted with probability $\frac{1}{2}$. Hence, the expected profit on a marginal limit order at an ask price below $2C + \mu$ is negative, so that no sell limit orders are submitted below $2C + \mu$. The best price on the sell side of the limit order book (the first price at which some shares are offered) is therefore the lowest price on the grid above $2C + \mu$. We denote this price by A^* .

What is the cumulative depth Y_k of the order book at price $A_k \geq A^*$? Consider a trader who contemplates placing a limit order at price A_k . Given the time-priority rule, this order goes to the back of queue Y_k and its execution probability is therefore at most $P(Y_k)$. The trader faces a trade-off between the revenue in case of execution of the limit order, which is $A_k - \mu$ per share, and the order submission cost C . The optimal point along this trade-off depends on the order's execution probability. If the queue Y_k at price A_k is short enough, then the execution probability is high enough so that the trader can expect a positive profit, that is, $\Pi_k(Y_k) \geq 0$. Otherwise, it is unprofitable for the trader to add a limit order to the queue at price A_k , that is, $\Pi_k(Y_k) < 0$.

Thus, the no-entry/no-exit condition for a competitive equilibrium holds when the cumulative depth at price $A_k \geq A^*$ solves:

$$\Pi_k(Y_k) = 0, \quad (6.5)$$

so that the probability of execution of the marginal limit order at price A_k is:

$$P(Y_k) = \frac{C}{A_k - \mu}, \quad (6.6)$$

$$\text{or } A_k = \mu + \frac{C}{P(Y_k)}. \quad (6.7)$$

Example 1. Consider the case in which the distribution of market order size is exponential, as in equation (6.1). In this case:

$$P(Y_k) = \frac{1}{2} e^{-\theta Y_k}.$$

Thus, solving equation (6.6) for Y_k , we obtain the cumulative depth at price $A_k \geq A^*$:

$$Y_k = \frac{1}{\theta} \ln \left(\frac{A_k - \mu}{2C} \right).$$

The determinants of cumulative depth. In the model, the cumulative depth at price A_k depends on the order submission cost and the probability distribution of market order size. This is clearly shown in example 1. The cumulative depth increases with $1/\theta$, the average order size, because people anticipate “the queue to go faster.” Further, cumulative depth at each price is decreasing in the order display cost C . Intuitively, the queue at or below price A_k is shorter when the cost of joining the line is higher. This means that an order entry fee for limit orders impairs the depth of the LOB at every price.

Empirically, the level of trading activity varies across stocks. For instance, firms with a large market value of equity, known as large caps (*LC*), are typically more active, in the sense that trades are more frequent and larger than for small caps (*SC*). This means that the likelihood of a market order larger than any given size is greater for large caps, i.e., $P_{LC}(Y) > P_{SC}(Y)$ for all $Y > 0$. Thus, by equation (6.6), the cumulative depth in the large-cap market in equilibrium will be greater at all prices. In this sense, liquidity demand (active trading) begets liquidity supply, as it improves the chance of execution for limit order traders and so induces them to submit more orders.

6.2.3 Limit Order Trading with Informed Investors

We now explore how adverse selection affects the LOB. We assume that market orders are submitted either by uninformed traders or by informed traders (those who know the true value of the security, $v = \mu + \varepsilon$). Limit order traders do not observe ε but they are aware that market orders may come from informed traders. This model is originally due to Glosten (1994).

Exposure to informed trading has two distinct sources. First, informed traders may have private information not yet available to the market, as Chapters 3 and 4 suggest. Second, as noted in the introduction to this chapter, the arrival of public information exposes limit order traders to the risk of being “picked off” if they are slow in updating their prices to reflect news. In each case, the actions of the informed traders create a positive correlation between the order flow and the change in the estimate of the true value, which is the essence of the adverse selection problem.

The very fact that a limit order is executed contains information. It reveals the direction of the market order and it shows that it was large enough to trigger execution of the limit order. This information is value-relevant, since the size and direction of the market order are correlated with the true value. So when placing a limit order and choosing one’s bid, one must take this information into account. Recall that, by equation (6.3), the expected profit on the marginal unit offered at price A_k is:

$$\Pi_k(Y_k) = P(Y_k)(A_k - E(v|q \geq Y_k)) - C, \quad (6.8)$$

where the upper tail conditional expectation, $E(v|q \geq Y_k)$, is the estimate of the value of the security conditional on execution of the marginal unit at price A_k .

Intuitively, with informed trading this upper tail expectation increases with Y_k , the cumulative quantity at each price. Consider an informed trader’s optimal strategy when he knows that the value of the security is $v > A_k$. If he is not wealth-constrained, his demand is unlimited at any price below v and zero at any price above. Thus, he will definitely sweep up all sell limit orders in the book at prices at or below A_k . In contrast, an uninformed trader only hits the limit orders needed to fill his desired order. Thus, as the queue of limit orders at price A_k lengthens, it becomes more and more likely that the market order triggering execution of the marginal order is from an informed investor who knows that the limit order is “stale,” i.e., that $v > A_k$. As a consequence, the marginal trader’s valuation conditional on execution gets larger as Y_k increases. The next example illustrates this point.

Example 2. The final value of the security can be either low ($v^L = \mu - \sigma$) or high ($v^H = \mu + \sigma$), with equal probability. The market order trader is either informed with probability π or uninformed with probability $1 - \pi$. The uninformed trader is a buyer or a seller with equal probability, with a desired trade size that is either small (q_s) or large (q_L) with probabilities ϕ and $1 - \phi$, respectively. Let $A(q_s)$ and $A(q_L)$ be the ask prices at which cumulative depth on the ask side is equal to q_s and q_L shares, respectively. Moreover, assume that

$A(q_L) < v^H$.² Thus, if the informed investor knows that the value of the security is v^H , he sweeps all limit orders up to price $A(q_L)$, at least.

Let I and U denote the event that the market order is submitted by the informed and the uninformed traders, respectively. Given the order submission strategy of liquidity demanders, we have:

$$\Pr(I|q \geq Y) = \frac{\Pr(q \geq Y|I) \Pr(I)}{\Pr(q \geq Y)} = \frac{\frac{\pi}{2}}{\frac{\pi}{2} + \frac{1-\pi}{2}} = \pi \text{ for } Y \leq q_s.$$

Indeed, an investor is informed with probability $\Pr(I) = \pi$ and, as explained previously, an informed investor buys at least $q_L > q_s$ shares when she learns that the final value is high, that is, $\Pr(q \geq Y|I) = \frac{1}{2}$ for $Y \leq q_s$. As an uninformed investor buys at least q_s shares with probability $\frac{1}{2}$, it follows that $\Pr(q \geq Y) = \frac{1}{2}$ for $Y \leq q_s$ as well. Following the same reasoning, we obtain:

$$\Pr(I|q \geq Y) = \frac{\Pr(q \geq Y|I) \Pr(I)}{\Pr(q \geq Y)} = \frac{\frac{\pi}{2}}{\frac{\pi}{2} + \frac{(1-\pi)(1-\phi)}{2}} = \frac{\pi}{\pi + (1-\pi)(1-\phi)}$$

for $q_s < Y \leq q_L$, because an uninformed investor only buys a large number of shares with probability $\frac{1-\phi}{2}$. It follows that the second expression is greater than the first. Intuitively, a trade smaller than q_s conveys no information about the identity of the person placing the market order, but a trade larger than q_s is more likely to come from an informed trader. It follows that:

$$\begin{aligned} E(v|q \geq Y) &= \Pr(U|q \geq Y)\mu + \Pr(I|q \geq Y)(\mu + \sigma) \\ &= \mu + \Pr(I|q \geq Y)\sigma. \end{aligned} \quad (6.9)$$

Hence $E(v|q \geq Y)$ is weakly increasing in Y because a limit order far back in the queue is more likely to be hit by an informed investor ($\Pr(I|q \geq Y)$ increases in Y). In particular,

$$E(v|q \geq q_s) = \mu + \pi\sigma, \quad (6.10)$$

and:

$$E(v|q \geq q_L) = \mu + \frac{\pi\sigma}{\pi + (1-\pi)(1-\phi)} > E(v|q \geq q_s). \quad (6.11)$$

² This will in fact be the case in equilibrium.

Let us now go back to deriving the equilibrium LOB. Since $E(v|q \geq Y_k)$ increases with Y_k , the expected profit on the marginal order at a given price decreases along with cumulative depth at this price. As in the previous section, a competitive equilibrium is obtained when cumulative depth Y_k at each price A_k is such that the following zero profit condition is satisfied:

$$\Pi_k(Y_k) = 0 \text{ if } \Pi_k(Y_{k-1}) > 0, \text{ and } Y_k = Y_{k-1} \text{ if } \Pi_k(Y_{k-1}) \leq 0, \quad (6.12)$$

meaning that the marginal unit must yield no profit if offering less were to yield positive profits; otherwise, that unit must not be offered. From equation (6.8), the zero profit condition is equivalent to:

$$A_k = E(v|q \geq Y_k) + \frac{C}{P(Y_k)} \text{ if } Y_k > Y_{k-1}, \quad (6.13)$$

$$A_k < E(v|q \geq Y_{k-1}) + \frac{C}{P(Y_{k-1})} \text{ if } Y_k = Y_{k-1}. \quad (6.14)$$

In the remainder of this section, in order to isolate the impact of asymmetric information on the equilibrium price schedule, we assume that the order submission cost C is zero. The example below illustrates, in a simple way, some important and general properties of the limit order market with informed trading.

Example 1, continued. Assume again, as in example 1, that the market order size has an exponential distribution $f(q) = \frac{1}{2}\theta e^{-\theta|q|}$ and that the expected value of the security conditional on the market order size is linear:

$$E(v|q = x) = \mu + \lambda x. \quad (6.15)$$

Equation (6.15) is the “updating rule” that specifies how liquidity suppliers (limit orders traders) would update their estimate of the asset value if they knew the total size of the market order. The parameter λ is therefore a measure of the informativeness of the order flow. In this example, we take the updating rule and the order size distribution as primitives. This parametric specification of the model is a shortcut for analyzing the properties of the LOB with asymmetric information and has proved useful empirically. Later on, we consider a richer setting in which the updating rule and the distribution of order size are endogenous.

Using the law of iterated expectations and the updating rule, we obtain:

$$\begin{aligned} E(v|q \geq x) &= E(E(v|q = x)|q \geq x) \\ &= \mu + \lambda E(q|q \geq x) \end{aligned}$$

$$\begin{aligned}
&= \mu + \lambda \frac{\int_x^\infty q \theta e^{-\theta q} dq}{\int_x^\infty \theta e^{-\theta q} dq} \\
&= \mu + \frac{\lambda}{\theta} + \lambda x, \text{ for } x \geq 0.
\end{aligned}$$

Consider first the case of zero tick size (i.e., prices are not required to take discrete values on a grid). In this case, from equation (6.13) the equilibrium price schedule on the ask side of the book is:

$$A(Y) = E(v|q \geq Y) = \mu + \frac{\lambda}{\theta} + \lambda Y.$$

This price schedule is shown as the upward-sloping line in Figure 6.5. Note that the relationship between bid prices and cumulative depth at each price is the mirror image of the ask side, since the probability distribution for market order sizes is symmetric around zero.

When the tick is not zero, so that prices must be at discrete intervals (no less than ten cents, say), the cumulative depth at each price on the ask side can be derived using equations (6.13) and (6.14). This yields:

$$Y_k = 0 \text{ for } A_k \leq \mu + \frac{\lambda}{\theta} \quad (6.16)$$

$$Y_k = \frac{A_k - (\mu + \frac{\lambda}{\theta})}{\lambda} \text{ for } A_k > \mu + \frac{\lambda}{\theta}. \quad (6.17)$$

Recall that we denote by A^* the lowest price in the limit order book at which some shares are supplied (the best offer). By definition, this price is the first price on the grid at which $Y_k > 0$ and therefore below this price the cumulative

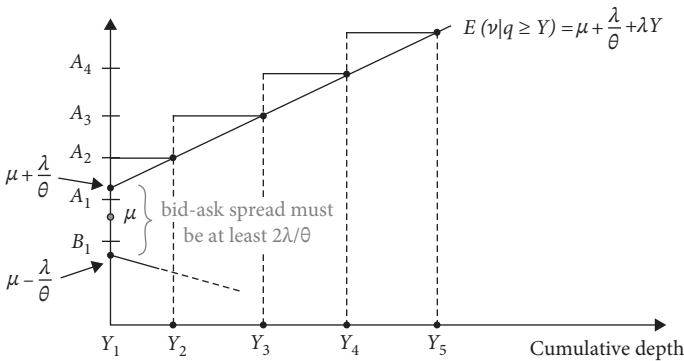


Figure 6.5 Equilibrium limit order book in example 1.

depth is zero. Thus, A^* is the first price on the grid above $E(v|q \geq 0)$, which in this case is $\mu + \frac{\lambda}{\theta}$. The number of shares offered at this price (the quoted depth) is $\frac{A^* - \mu}{\lambda} + \frac{1}{\theta}$. Graphically, the ask side of the book in this case is represented by the step function in Figure 6.5. The equilibrium has two interesting properties:

- **The limit order market features a non-zero bid-ask spread even for very small orders.** That is, even market orders for an infinitesimal quantity get executed at a premium or discount relative to the expected value of the security. This feature is not an effect of the tick size, since it persists even when the tick is zero, in which case the spread for a very small order is $\frac{2\lambda}{\theta} > 0$. This property does not obtain in the models presented in Chapter 4, such as that of Kyle (1985), in which the spread on an order of size q is given by $2\lambda q$ and goes to zero as the order size goes to zero. The reason is that in the limit order market, liquidity suppliers cannot make their quotes contingent on the total size of market orders. As a result, the limit orders at the top of the book are more exposed to adverse selection. To see this, let us take the sell limit order with the most competitive ask price. It will execute against market buy orders of every size, as it takes precedence over all the other limit orders. So in choosing his offer, the trader at the top of the book must take account of the possibility that his offer can be hit both by very small market orders (which convey very little information) and by very large ones (which are strongly informative). For this reason, his valuation conditional on execution is strictly larger than his prior estimate of the asset value μ .³
- **The depth of the LOB is decreasing in the informativeness of order flow.** As λ increases, the LOB thins out (i.e., the cumulative depth at each price diminishes). The intuition should by now be familiar to the reader: the more informative the order flow, the more exposed liquidity suppliers are to adverse selection, so they bid less aggressively (see Chapter 4).

In this example, we have taken the distribution of market order sizes and the updating rule as given. This approach simplifies the analysis but is not entirely satisfactory, as these functions ultimately depend on the informed trader's order placement strategy, which itself depends on the offers available in the book. Thus, in equilibrium the informed investor's order placement strategy and the LOB are

³ The trader's estimate of the value of the security conditional on execution is $E(v|q \geq 0) = \mu + \frac{\lambda}{\theta} > \mu$.

determined *jointly*. We now provide a new example in which this is the case, to show the robustness of the conclusions obtained so far (see also exercise 1).

Example 2, continued. Consider example 2 again and assume that the tick size is zero. In this case, the equilibrium is as follows. If the investor arriving at time 1 is informed, he submits an order of q_L shares, and the LOB displays the following offers:

$$A(Y) = E(v|q \geq Y) = \begin{cases} \mu + \pi\sigma & \text{for } Y \leq q_S, \\ \mu + \frac{\pi}{\pi + (1-\pi)(1-\phi)}\sigma & \text{for } q_S < Y \leq q_L, \\ \mu + \sigma & \text{for } Y > q_L. \end{cases} \quad (6.18)$$

$$B(Y) = E(v|q \leq -Y) = \begin{cases} \mu - \pi\sigma & \text{for } Y \leq q_S, \\ \mu - \frac{\pi}{\pi + (1-\pi)(1-\phi)}\sigma & \text{for } q_S < Y \leq q_L, \\ \mu - \sigma & \text{for } Y > q_L. \end{cases} \quad (6.19)$$

To verify that this is an equilibrium, we must show that the informed investor's behavior is optimal, given the offers in the book, and that the LOB is in a competitive equilibrium, given the behavior of the informed investor.

The first step is straightforward. Suppose that the investor arriving at time 1 is informed and learns that the value of the security is high. He can buy up to q_L shares at a price strictly below the value $v^H = \mu + \sigma$. Thus, buying at least q_L shares is optimal for this informed investor. Additional shares are purchased exactly at v^H . Hence, buying q_L shares is optimal for the informed trader.

Given this behavior, we can follow exactly the same reasoning as in example 2 to show that $E(v|q \geq Y)$ is given by equation (6.9) for $Y \leq q_L$. Moreover, only the informed trader submits a buy market order larger than q_L and he does so upon learning that the true value is v^H . That is,

$$E(v|q \geq Y) = v^H = \mu + \sigma \text{ for } Y > q_L.$$

Thus, in a competitive equilibrium, ask prices in the book are as given by equation (6.18). The analysis for the bid side is symmetric.

This example confirms the robustness of the conclusions drawn from the previous example. First, the book becomes thinner as the exposure to informed trading (π) heightens. And even though the informed investor never submits a small market order, these orders execute at a markup relative to μ . Again, this reflects the fact that limit orders at the top of the book may execute against small orders (with no information content) or large orders (which are informative), leading to a non-zero bid-ask spread for small trades:

$$A(q) - B(q) = 2\pi\sigma, \quad \text{for } 0 \leq q \leq q_S.$$

6.3 The Design of LOB Markets

The model developed in the previous section can be used to address market design issues: how liquidity and the distribution of trading gains among traders are affected by tick size (Section 6.3.1), secondary priority rules (Section 6.3.2), and designated market makers like the NYSE specialist (Section 6.3.3).

6.3.1 Tick Size

Market organizers or regulators usually set a tick size that varies across exchanges and across stocks, as it often is a function of prices. On the NYSE, it was $\frac{1}{8}$ for stocks with prices over one dollar until June 1997, when, under regulatory pressure, it was reduced to $\frac{1}{16}$ and finally to one cent in 2000. Decimalization was imposed on Nasdaq and AMEX as well. This change was very controversial and the appropriate size of the tick is still debated both in the U.S. and in Europe. For instance, the JOBS Act (2012) required the SEC to conduct a study on “how decimalization affected the number of initial public offerings (IPOs) and the liquidity and trading of smaller capitalization company securities.” Following this request, the SEC implemented a pilot program in 2016 to study the effect of raising the tick size to 5 cents for some small-cap stocks.

It may seem surprising that investors, exchanges, intermediaries, and policy makers care so much about tick size. After all, it is often very small compared to stock prices. Yet as the model shows, the tick determines liquidity suppliers’ total expected profits. In the model, the number of shares offered at each price is such that the expected profit on the last share (the marginal limit order) is just zero. However, inframarginal limit orders (those ahead of the marginal order in the queue at a given price) obtain strictly positive expected profits *if* the tick size is strictly positive. Consider for instance the trader at the head of the queue of limit orders at price $A_k > A^*$ (i.e., a price strictly above the best offer in the limit order book). His expected profit on the first share offered is:

$$\Pi_k(Y_{k-1}) = (1 - F(Y_{k-1}))(A_k - E(v|q \geq Y_{k-1})) - C, \quad (6.20)$$

because this order is filled if and only if the size of the market order exceeds the cumulative depth offered at price A_{k-1} . In a competitive equilibrium, the zero profit condition on the marginal share at price A_{k-1} requires:

$$A_{k-1} = E(v|q \geq Y_{k-1}) + \frac{C}{(1 - F(Y_{k-1}))}. \quad (6.21)$$

Equations (6.20) and (6.21) imply:

$$\Pi_k(Y_{k-1}) = (1 - F(Y_{k-1}))(A_k - A_{k-1}) = (1 - F(Y_{k-1}))\Delta > 0.$$

Thus, the expected profit on the “first” share sold at A_k is proportional to the tick size. The expected profit on additional shares offered at A_k is lower because their probability of execution is lower. Thus, the aggregate expected profit of all limit orders submitted at A_k decreases when the tick size declines. This is also the case for limit orders submitted at the best offer price.⁴ In the extreme case of a zero tick size, limit order traders get zero expected profits. Otherwise, their aggregate expected profit is strictly positive and bounded by the tick size.

Consistent with this mechanism, Yao and Ye (2018) empirically find that liquidity providers’ rents are higher for stocks with larger relative tick sizes and that they fiercely compete to be at the head of the queue of limit orders at a given price. Indeed, time priority implies that these traders obtain larger expected profits than those at the back of the queue.

The tick size is a source of profit for liquidity providers because it hinders competition: it stops them from bidding the security price up or down to their marginal valuation. As we have just seen, the marginal valuation for the first share offered at price $A_k > A^*$ is A_{k-1} , which is less than A_k if the tick size is positive. Traders could thus earn a strictly positive expected profit by undercutting this offer by an infinitesimal amount, but given the tick, they cannot. If they undercut a quote, they must do so by at least Δ . In equilibrium, the depth at each price is just enough to make undercutting unprofitable.

Figure 6.6 shows the effects of cutting the tick in half from Δ to $\frac{\Delta}{2}$ in the context of example 1. This cut creates new eligible prices on the grid,² which are referred to as “new” in Figure 6.6 (other prices are eligible in both grids). The graph shows that the reduction in the tick size can reduce the best offer and therefore the quoted bid-ask spread. When this happens, a reduction in the tick size can reduce the number of shares offered at the best quotes (the quoted depth), as in Figure 6.6. Moreover, the number of shares posted at any given price on the old grid is reduced as well. Intuitively, some of the shares formerly offered at any price A_k are now offered at price $A_k - \frac{\Delta}{2}$. However, the

⁴ Indeed, the best offer is the first price on the grid above $E(v|q \geq 0) + 2C$. Therefore the expected profit on the first share offered in the LOB is the difference between this price and $(E(v|q \geq 0) + 2C)$. Therefore, as the tick size declines, the aggregate expected profit of limit orders submitted at the best offer declines.

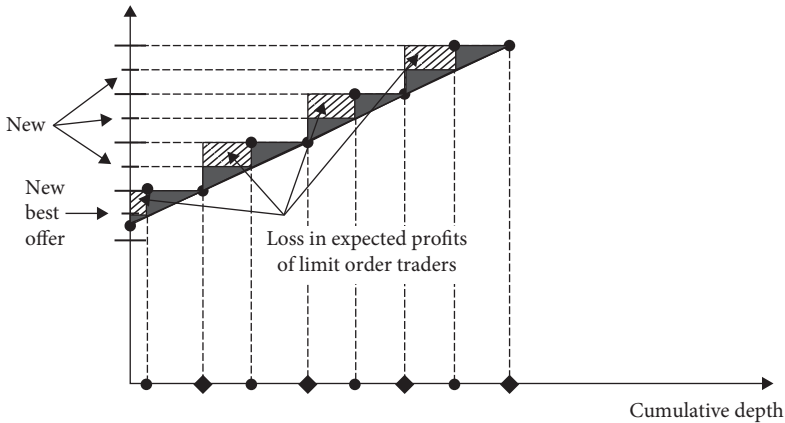


Figure 6.6 Effect of a reduction in the tick size. The dots (diamonds) on the x axis represents the cumulative depth at each new (old) offer price (y axis) on the grid. The dashed areas represent the loss in expected profits for liquidity suppliers following the reduction in the tick size.

smaller tick does not affect the cumulative depth at a given price on the old grid, because cumulative depth at each price must adjust so that the valuation of the marginal seller just equals the price. If this valuation is unaffected by the change in the tick size (as assumed here), then the cumulative depth *at a given price* is independent of the tick size. Thus, reducing the tick size unambiguously lowers the total trading cost for market orders in the framework considered here.

For instance, consider a buy market order of size q . Its total purchase cost is given by the area under the step function $A(q)$, mapping cumulative depth and quotes in equilibrium. Clearly this area becomes smaller when the tick size is reduced. The decrease in trading costs for liquidity demanders reflects the fact that the smaller tick intensifies competition among liquidity suppliers. For instance, the offers posted at price $A_k - \frac{\Delta}{2}$ compete away some of the profits at price A_k on the old grid and so reduce total trading costs.

To sum up, the model predicts that a decrease in tick size:

- (i) increases price competition (undercutting) between liquidity suppliers;
- (ii) reduces expected profits of liquidity suppliers and therefore trading costs for liquidity demanders;
- (iii) reduces the number of shares offered at each price, leaving cumulative depth at a given price unchanged;
- (iv) reduces the bid-ask spread;

These points explain why reducing the tick size is such a controversial measure. It clearly hurts the “sell-side” firms that supply liquidity (e.g., firms with market-making activities), and it benefits the “buy side” that demands liquidity. Not surprisingly, the two sides disagree over reducing the tick.

There is evidence that a finer price grid encourages price competition. For instance, Foley et al. (2023) study a (temporary) reduction in tick size in 2009 by Chi-X, BATS, and Turquoise (three European equity trading platforms) for stocks listed on the London Stock Exchange and various Nordic Exchanges (Denmark, Norway, and Sweden). They find that liquidity providers used the finer grids on these platforms to undercut quotes posted on the main markets for these stocks (which were using a coarser grid). They also show that as a result trading costs for these stocks declined and consolidated depth increased.

Other empirical studies find that changes in tick size have different effects on the liquidity available to small and large traders. For instance, Chung et al. (2020) study the effect of an increase in the tick size for small cap stocks that are part of the SEC (2016) pilot program (“pilot stocks”), and find that this increase raises trading costs for small orders, but decreases trading costs for large orders (between five hundred and five thousand shares) due to an increase in cumulative depth throughout the consolidated limit order book of pilot stocks. Interestingly, these findings mirror those reported by Goldstein and Kavajecz (2000) who studied the effect of the reduction in the tick size from $\$ \frac{1}{8}$ to $\$ \frac{1}{16}$ on the NYSE in 1997.⁵

Overall, these empirical findings paint a more complex picture than is suggested by the simple model analyzed in this section. They confirm the prediction that a decrease in the tick size should reduce bid-ask spreads and trading costs for small orders. However, they also suggest that such a decrease can increase trading costs for large orders, in contrast to what the model predicts. This may be due to the fact that the model ignores several possible side-effects. First, as a non-zero tick size guarantees minimal profits for liquidity providers, reducing it may prompt them to exit the market and so reduce total liquidity provision. Moreover, limit order traders may pay less attention to the LOB as the reduction in the tick size diminishes the gain from being at the head of the queue. Or they may choose to improve non-competitive offers in the book by a smaller amount rather than directly posting a competitive offer (since the expected profit obtained at the competitive offer is smaller). In both cases, it will take longer for

⁵ Bacidore (1997), Porter and Weaver (1997), and Ahn, Bae, and Kalok (1998) studied the impact of a *reduction* in the tick size on the Toronto Stock Exchange in April 1996. They found a reduction of 17.27% in the quoted spread and of 27.52% in the quoted depth at the best price (depending on study and sample).

the LOB to be replenished after a trade, and market orders will execute more frequently against non-competitive offers.⁶ Lastly, if the return to limit order trading is lower, traders might use market orders more frequently. Again, this effect could decrease liquidity when the tick size is reduced.

6.3.2 Priority Rules

The LOB can be seen as a queue. Priority rules determine the sequence in which the orders in the queue are “served.” Price priority comes first. That is, limit orders at a given price are executed before those at worse prices. Secondary priority rules determine the sequence of execution of multiple limit orders at any given price. The most common is time priority. That is, limit orders at the same limit price are eligible for execution in the sequence in which they were submitted. But there are also other rules. For instance, the electronic futures markets for the leading short-term interest rate in the United States (Eurodollar), Europe (Euribor), and the United Kingdom (Short Sterling) and for the two-year U.S. Treasury Note future have a pro-rata allocation rule.

To see how this rule operates, consider two sell limit orders tied at the best ask price, the first offering one thousand shares, and the second four thousand shares. Suppose a buy market order for four thousand shares arrives. Under the pro-rata rule, the first limit order serves 20% of the incoming order (800 shares) and the second serves the remaining 80% (3,200 shares). Other tie-breaking rules can also be considered. For instance, the market order could be assigned randomly to all the limit orders at the same price.

Priority rules affect traders’ bidding behavior and thereby market liquidity. To see this, we compare the LOB with time priority and that with pro-rata allocation. Let Y_k^r be the cumulative depth at price A_k under the pro-rata allocation rule. Using equation (6.8), the aggregate expected profit of all the limit orders posted at price A_k is:

$$\int_{Y_{k-1}^r}^{Y_k^r} \Pi_k(y) dy = \int_{Y_{k-1}^r}^{Y_k^r} \{P(y) [A_k - E(v|q \geq y)] - C\} dy. \quad (6.22)$$

Under the pro-rata rule, this expected profit is shared among all the investors with a limit order at price A_k in proportion to the size of each one’s order. Thus,

⁶ This effect is analyzed in Cordella and Foucault (1999) and Foucault, Kadan, and Kandel (2005). Both studies conclude that the tick size that minimizes trading costs for liquidity demanders is strictly positive.

a competitive equilibrium is reached when the cumulative depth at each price in the LOB satisfies:

$$\int_{Y_{k-1}^r}^{Y_k^r} \Pi_k(y) dy = 0 \quad \forall k. \quad (6.23)$$

Otherwise, it would be profitable either to add limit orders to the book (to get a fraction of the aggregate expected profit at price A_k) or to withdraw limit orders from the book (if the aggregate expected profit is negative).

As $\Pi_k(\cdot)$ is a decreasing function, the zero profit condition (6.23) implies that $\Pi_k(Y_k^r) < 0$. But recall that under time priority, the cumulative depth at price A_k , which we denote here by Y_k^t , satisfies $\Pi_k(Y_k^t) = 0$. Hence,

$$Y_k^r > Y_k^t. \quad (6.24)$$

Thus, the market is always deeper under pro-rata allocation, holding the distribution of market order flow constant (see exercise 3).

This result assumes that the probability distribution of the market order size does not change when the priority rule is changed. Actually, liquidity demanders are likely to submit larger orders when the book is deeper. In Section 6.2.2 such a shift in the distribution of uninformed market orders results in larger cumulative depth. Thus, with an endogenous distribution for market order size, the positive effect of the pro-rata allocation rule on market depth would be even stronger. However, the effect is less clear-cut with informed trading, since informed traders' strategies and the depth of the LOB are jointly determined (see Section 6.2.3). Thus, the net liquidity effect of a change in the priority rule (or, more generally, any change in market design) ultimately requires an analysis of the reaction of both limit order traders and informed traders. Exercise 2 provides an example.

Last, one may wonder why time priority is used in real markets if the pro-rata allocation rule results in a more liquid LOB. As with tick size, a broader perspective is required. In a dynamic setting, time priority rewards liquidity suppliers who are quick to replenish the book when it is depleted. This effect is not considered in the model. Also, exchanges may want to guarantee some minimal profit for limit order traders to encourage liquidity provision. For instance, some exchanges give rebates to limit order traders in case of execution, which shows that they see the profitability of limit orders as critical to their success. Indeed, if the profitability of a limit order is too low compared to its cost or the potential gain from alternative strategies (e.g., placing market orders), the liquidity of the LOB could dry up, endangering the viability of the limit order market.

6.3.3 Hybrid LOB Markets

In limit order markets, trading platforms sometimes assign one or several market makers to a stock. In this case, liquidity is provided both by the LOB and by the designated market makers' offers. In some exchanges (like Euronext), the designated market makers simply post additional limit orders in the LOB, under the constraint that their spread not exceed a threshold set by contract with the exchange or the issuer.⁷ In other exchanges, like the NYSE or the Frankfurt Stock Exchange, designated market makers were allowed to make offers contingent on the size of incoming market orders ("last mover advantage"). That is, instead of providing liquidity *ex ante* (before observing the size of the incoming market orders) like regular limit order traders, they could also do so *ex post*, i.e., after observing the size of the orders). The NYSE suppressed this arrangement in 2008. However, as explained at the end of this section, this possibility is often implicitly or explicitly allowed by some trading platforms (see Box 6.1).

Is granting such a last mover advantage a good idea? To study this, we extend the model developed in Section 6.2.3 as described in Figure 6.7. After the arrival of a market order, one market maker—the "specialist"—chooses a "stop-out price" $p^s(q)$ at which he is willing to execute all or part of the market order.⁸ This price depends on the direction and size of the market order, since unlike limit order traders the specialist sets his stop-out price *after* the market order arrives.

We assume that the specialist must yield both price and time priority to the book (as was the case on the NYSE until 2008). Say the LOB displays one thousand shares at $A_1 = 100$ and six hundred shares at $A_2 = 100.1$. A buy market

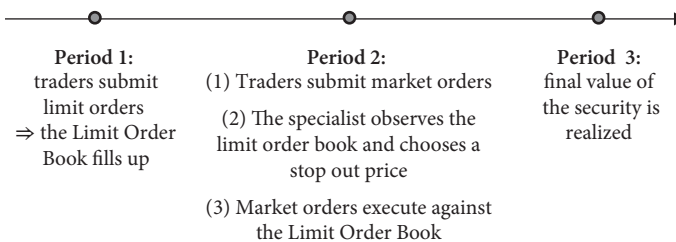


Figure 6.7 Timing with a specialist granted a last mover advantage.

⁷ In compensation, the designated market makers receive a payment from the issuer or are exempted from the exchange's trading fees. We discuss further the role of designated market makers in Section 1.2.3 and in Section 10.5.2.

⁸ We refer to the market maker as the "specialist" because this used to be the term used to refer to the market maker assigned to one stock on the NYSE. The NYSE specialist for a stock had the last mover advantage that we analyze in this section. See, for instance, Seppi (1997).

order for two thousand shares arrives and the specialist decides to “stop” the order out at $p^s(2000) = 100.1$. In this case, the buy order executes first against the limit orders at $A_1 = 100$ and $A_2 = 100.1$, after which the specialist fills the order for the remaining four hundred shares at $p^s = 100.1$.

At first glance, it seems that the specialist’s intervention could only enhance market liquidity since he expands that offered by the LOB. But this thesis neglects the impact of the specialist on the bids of limit order traders. In the presence of asymmetric information, the specialist’s ability to condition his intervention on the exact size of the market order allows him to cream-skim orders with little information content (small orders). This last-mover advantage reinforces the exposure of limit orders to informed trading and so impairs liquidity.

To see this, let us revisit example 2 in Section 6.2.3. As noted, in the equilibrium described in this example, small market orders (of size q_s) are submitted only by uninformed investors. The specialist can exploit this in the following way. When a small order arrives, he stops it out by slightly undercutting the prices offered by limit order traders for an order of this size, that is, at $A(q_s)$ or $B(q_s)$ (respectively given by equations (6.18) and (6.19)) depending on the direction of the order; when a large order arrives, he refrains from trading. This strategy yields a positive expected profit amounting to:

$$\frac{(1-\pi)\phi}{2} (A(q_s) - \mu) + \frac{(1-\pi)\phi}{2} (\mu - B(q_s)) = (1-\pi)\pi\phi\sigma > 0.$$

Intuitively, by cherry-picking the small orders, the specialist captures the compensation required by limit order traders for their exposure to informed trading without being exposed himself. The specialist can do this because, unlike limit order traders, he can condition his intervention on the exact size of the market order.

But then the equilibrium described in example 2 unravels. Indeed, in this case, limit orders at the top of the book execute if and only if a market order for at least q_L shares arrives. Hence, they no longer break even at price $A(q_s)$. Instead, the equilibrium of the hybrid market with a last mover advantage can be readily determined as follows. If the investor arriving at time 1 is informed, he submits an order of at least q_L shares and the LOB displays the following prices (where the superscript h stands for “hybrid”):

$$A^h(Y) = \begin{cases} \mu + \frac{\pi}{\pi + (1-\pi)(1-\phi)}\sigma & \text{for } Y \leq q_L, \\ \mu + \sigma & \text{for } Y > q_L, \end{cases} \quad (6.25)$$

$$B^h(Y) = \begin{cases} \mu - \frac{\pi}{\pi + (1-\pi)(1-\phi)}\sigma & \text{for } Y \leq q_L, \\ \mu - \sigma & \text{for } Y > q_L. \end{cases} \quad (6.26)$$

Moreover, at time $t = 1$, the specialist stops small market orders out by slightly improving on the best quotes in the book ($A(q_L)$ or $B(q_L)$ depending on whether the market order is a buy or a sell). If a large market order arrives, the specialist stays put.

Thus in equilibrium the difference between the quotes at the top of the book now is:

$$A^h(0) - B^h(0) = \frac{2\pi\sigma}{\pi + (1-\pi)(1-\phi)} > A(q_s) - B(q_s) = 2\pi\sigma, \quad (6.27)$$

so that the quoted spread for small orders in the hybrid market is strictly larger than in the pure limit order market, i.e., $A(q_s) - B(q_s)$. Indeed, the specialist's intervention has a perverse effect. It increases limit order traders' exposure to informed trading. As the spread on large orders is identical in the two market structures, all liquidity demanders are worse off in the hybrid limit order market: in the pure limit order market, small and large market buy orders execute wholly or at least partially against the limit orders at $A(q_s)$, while in the hybrid, all market orders execute at a higher price, namely $A(q_L)$.

To sum up, the hybrid market with a last mover advantage aggravates the adverse selection problem for those without this advantage, and so reduces the liquidity of the LOB. In our analysis, only one liquidity provider (the specialist) benefits from the last mover advantage, which dramatically increases trading costs. Exercise 4 shows that competition among multiple liquidity providers alleviates the problem raised here, but does not eliminate it.

This problem is just an illustration of a more general principle. The possibility for some liquidity providers to post quotes contingent on the observation of some trader or order characteristic (here the size of the trade) enables them to cream-skim uninformed trades. Cream-skimming raises the exposure of other liquidity providers to adverse selection and ultimately impairs liquidity. This issue resurfaces regularly in debates about market design (see Box 6.1 for an example or the discussion on payment for order flow in Section 7.2.4 and in Section 8.3).

Box 6.1 Flash Orders

With the development of algorithmic trading and the intensification of competition for order flow, some electronic trading platforms (e.g., Nasdaq and BATS) introduced flash orders in 2009. In the United States, market orders must be executed at the national best bid and offer price (NBBO), i.e., the best of all competing trading venues for a stock. So a platform that receives

a buy market order and does not have the best ask price, must reroute the buy order to the platform that posts the best ask (see Section 7.5.1). If it is a flash order, though, before doing so the platform shows (“flashes”) the order to its participants at the national best offer for flash orders to buy, and the national best bid for flash orders to sell. Market participants that receive the flashed order information then have a very brief period (generally less than a second) to respond with their own order to execute against the flashed order at a price that matches the nationwide best quote. Flashed orders are a way for a platform to avoid losing order flow (and therefore revenues) to competing platforms.

As the permitted response time to flash orders is fleeting, only liquidity providers using algorithms have a chance to execute the flashed order. Thus, flash orders give a first look advantage to the traders who can actually respond to flashed orders. As a result, as the model developed in this section shows, such orders can increase the exposure of slow limit order traders (who do not react fast enough to respond to flashed orders) or those active in competing platforms to adverse selection and ultimately raise trading costs. In fact, this problem was pointed out by a number of market participants. For instance, after Nasdaq and BATS requested authorization from the SEC for flash orders, Morgan Stanley said the proposed rule changes “will provide a material disincentive to publicly display limit orders on exchanges, thereby impairing price discovery.”⁹ Given the controversies raised by flash orders and political pressure, both NASDAQ and BATS voluntarily discontinued their flash orders functionality on September 1, 2009.¹⁰

6.4 The Make or Take Decision in LOB Markets

A key feature of limit order markets is that traders can choose whether to act as makers or takers (either post or hit a limit order). These markets operate well when some investors are makers and others are takers. As we shall see here, the make or take decision hinges on the trade-off between the benefit of trading at a better price and the cost of being exposed to non-execution and pick-off risk. Section 6.4.1 formalizes this trade-off. Sections 6.4.2 and 6.4.3 investigate

⁹ SEC release no. 34 - 60684: “Elimination of Flash Order Exception from Rule 602 of Regulation NMS.”

¹⁰ Skjeltorp, Sojli, and Tham (2016) study empirically the impact of Nasdaq decision to introduce and then remove the flash order functionality. In contrast with the argument developed here, their findings suggest that flash orders improve market quality. They argue that this may be due to the fact that flash orders increased competition for order flow between limit order traders.

how the risk of non-execution and the risk of being picked off affect the bid-ask spread and its relationship with volatility. Section 6.4.5 analyzes how the state of the LOB (the number of shares offered on the bid and ask sides) affects traders' choices between market and limit orders.

6.4.1 Risk of Being Picked Off and Risk of Non-Execution

Consider the market for a security. Its final payoff v_T is realized at time T and is equal to:

$$v_T = \mu_0 + \sum_{t=1}^{t=T} \varepsilon_t, \quad (6.28)$$

where ε_t is a zero-mean innovation of which market participants are informed at the beginning of period t ; ε_t is equal to $+\sigma$ ("good news") or $-\sigma$ ("bad news") with equal probability. The final payoff time, T , is random. Specifically, in each period, there is a probability $1 - \tau$ that this payoff is realized, so that the market for the asset is closed. The parameter τ is an inverse measure of the participants' eagerness to clinch the trade: the closer τ is to zero, the more impatient they are.

At time t , if the asset has not yet paid off, its expected value—its "fair value"—is:

$$v_t = E_t(v_T) = \mu_0 + \sum_{k=1}^{k=t} \varepsilon_k = v_{t-1} + \varepsilon_t. \quad (6.29)$$

Thus, σ is the per-period volatility of the security's fair value.

After the realization of each ε_t , a new trader comes to the market. Traders have different valuations of the stock. The utility of trader i if he trades $q \in \{+1, 0, -1\}$ shares at price p is:

$$U(q, y_i, p) = q(v_T + y_i - p).$$

Parameter y_i is specific to each trader; it represents his "private value" from trading. It takes one of two values, $+L$ or $-L$, with equal probability. Intuitively, traders with a large gain from holding the security ($y_i = +L$) should buy from those with a low valuation ($y_i = -L$).¹¹

¹¹ Several stories come to mind for why traders have different valuations y_i . One is that traders have a long or short position in the security or in another asset whose payoff is correlated with v_T . In this case, traders with a long position value a purchase of the security less than traders with a short

The trader arriving at time t observes v_t and the state of the LOB. Given this information and his private valuation, the trader must choose whether to buy or sell, and whether to place a limit order or a market order. The market order executes immediately at the best quotes available.

In this setting, traders submitting limit orders are exposed to both the risk of being picked off (see Box 6.2) and the risk of non-execution. To see this, consider a trader who submits, at time t , a buy limit order at bid price B valid for only one period. If the trader is of type y_i , his expected utility with this order is:

$$E_t[U(q_{t+1}, y_i, B)] = E_t[(v_T + y_i - B) \cdot q_{t+1}], \quad (6.30)$$

where $q_{t+1} = 1$ if the order executes and $q_{t+1} = 0$ otherwise. Let $P_t(B)$ be the execution probability of this order. Then we can rewrite (6.30) as:

$$\begin{aligned} E_t[U(q_{t+1}, y_i, B)] &= E_t[(v_T - v_t + v_t + y_i - B) \cdot q_{t+1}] \\ &= \left[(v_t + y_i - B) + E\left(\sum_{k=t+1}^{k=T} \varepsilon_k | q_{t+1} = 1\right) \right] P_t(B) \\ &= [(v_t + y_i - B) + E(\varepsilon_{t+1} | q_{t+1} = 1)] P_t(B), \end{aligned} \quad (6.31)$$

where we have used the fact that $E\left(\sum_{k=t+2}^{k=T} \varepsilon_k | q_{t+1} = 1\right) = E\left(\sum_{k=t+2}^{k=T} \varepsilon_k\right) = 0$. Indeed, the decision of the trader arriving at date $t+1$ (i.e., q_{t+1}) cannot depend on the innovations in the asset value subsequent to date $t+1$ (ε_{t+2} etc....) because these are unknown to this trader. The first component $(v_t + y_i - B)$ is what the trader would gain with certain execution of his limit order if execution were uncorrelated with changes in the expected value of the security, so that $E(\varepsilon_{t+1} | q_{t+1} = 1) = E(\varepsilon_{t+1}) = 0$. His expected utility is less than this, however, because there is both a risk of non-execution ($P_t(B) < 1$) and of being picked off (the trader's buy limit order is more likely to be filled when $\varepsilon_{t+1} < 0$ than when $\varepsilon_{t+1} > 0$, so that $E(\varepsilon_{t+1} | q_{t+1} = 1) \leq 0$).

To illustrate these effects, consider a numerical example with $\sigma = 0.7$ and $L = 1$. A trader arrives at time t when $v_t = 100$. The LOB at time t has a single sell limit order at $A_t = 100.1$. If the trader submits a buy market order, his expected utility is:

position and conversely, as explained in Chapter 3. Or traders may have different discount factors or face different tax treatment and hence have different valuations for identical payoffs. For instance, for individual investors in the United States, interest on municipal bonds is tax-exempt while payouts of other securities are not; in contrast, pension funds are income tax-exempt irrespective of the securities they hold. Thus, other things being equal, individuals value munis more than tax-exempt institutions. Last, one could interpret y_i as reflecting differences of opinion among traders about the final payoff.

$$E_t[U(1, 1, 100.1)] = 101 - 100.1 = 0.9.$$

Now, suppose that instead he submits a buy limit order at $B = 99.9$. The expected utility from this order depends on whether it executes and on the evolution of the fair value of the security. There are four possible events at time $t + 1$:

1. Public information is good ($\varepsilon_{t+1} = +0.7$) and the trader arriving at time $t + 1$ submits a sell market order. The expected utility of the trader who arrived at time t is:

$$E_{t+1}[U(1, 1, 99.9)] = 100.7 + 1 - 99.9 = 1.8.$$

2. Public information is good ($\varepsilon_{t+1} = +0.7$) and the trader arriving at time $t + 1$ submits a limit order. In this case the trader who arrived at time t does not trade, obtaining expected utility of $E_{t+1}[U(0, 1, 99.9)] = 0$.
3. Public information is bad ($\varepsilon_{t+1} = -0.7$) and the trader arriving at time $t + 1$ submits a sell market order. The expected utility of the trader who arrived at time t is then:

$$E_{t+1}[U(1, 1, 99.9)] = 99.3 + 1 - 99.9 = 0.4.$$

4. Public information is bad ($\varepsilon_{t+1} = -0.7$) and the trader arriving at time $t + 1$ submits a limit order. The trader who arrived at time t does not trade and gets zero.

Let φ be the probability of an increase in the value conditional on execution, that is, the probability of event 1 relative to the total probability of execution (events 1 and 3). Then, at time t , the expected utility from the limit order is:

$$\begin{aligned} E_t[U(q_{t+1}, 1, 99.9)] &= \{(v_t + y_t - B) + E_t(\varepsilon_{t+1} | q_{t+1} = 1)\} P_t(B) \\ &= \{1.1 + [0.7\varphi - 0.7(1 - \varphi)]\} P_t(B) \\ &= [1.1 + 0.7(2\varphi - 1)] P_t(B) = (0.4 + 1.4\varphi) P_t(B), \end{aligned}$$

If $\varphi < 0.5$, we have $E(\varepsilon_{t+1} | q_{t+1} = 1) < 0$ because execution is more likely if the fair value goes down than if it goes up. This effect clearly reduces the expected gain from the limit order. Thus, the lower is φ , the higher is the investor's risk of being picked off.

The limit order at 99.9 dominates a market order if:

$$E_t[U(q_{t+1}, 1, 99.9)] \geq 0.9.$$

Thus, the decision on order type depends on φ and $P_t(B)$. For instance, if $P_t(B)$ is large and φ is close to $\frac{1}{2}$, then a limit order at 99.9 dominates the market order. But suppose instead that the limit order is only executed when the news is bad at time $t+1$ and that the next trader has a low private valuation: $P_t(B) = 0.25$ and $\varphi = 0$. In this case, the trader's expected utility with a limit order at 99.9 is:

$$E_t[U(q_{t+1}, y_t, B)] = (1.1 - 0.7) \cdot 0.25 = 0.1,$$

implying that he would be better off with a market order.

Of course, there may be another limit order at a different price that yields a larger expected utility, so that eventually the trader will submit a limit order. Finding the optimal limit order requires knowledge of the execution probability and the probability φ of good news conditional on execution associated with each possible limit order. These probabilities cannot be arbitrarily specified, since they depend on future traders' order choices in each possible future state of the market (defined at any given time by the type of trader arriving, the fair value of the security, and the offers present in the LOB). In a rational expectations equilibrium, the probabilities must be consistent with the order submission strategies that are used in equilibrium.

For this reason, it is difficult to solve for the equilibrium when traders can choose between market and limit orders.¹² In particular, the number of possible future states is very large if limit orders can stay on the LOB for many periods. To simplify the analysis, let's assume that all limit orders expire after one period, as in the last example. In Sections 6.4.2 and 6.4.3, we show how traders' optimal order submission strategies can be derived in this case. As we shall see, this analysis affords insights into the impact of execution risk and volatility on liquidity in limit order markets.

Box 6.2 SOES Bandits

The risk of being picked off stems from the fact that traders do not monitor their limit orders continuously and/or react too slowly to the arrival of value-relevant news. Hence, they may be too slow in updating their quotes when new information emerges. Market makers too are exposed to this risk. The controversy in the 1990s over Nasdaq's small order execution system (SOES) provides a good illustration. SOES was introduced by Nasdaq to provide automatic execution of small orders (say, less than five hundred shares)

¹² Another reason why the problem is difficult is that investors submitting limit orders may also want to revise them as market conditions change.

against dealers' quotes. Hence SOES also allowed traders who were quick in reacting to news to pick off Nasdaq dealers who were slow to adjust their quotes. By contrast, for larger orders, dealers had to first confirm that they were willing to execute, which gave them time to first check that their quotes were in line with current available information (e.g., other dealers' quotes). Thus, although initially intended for retail investors, SOES quickly became a trading tool for professional traders. As they were inflicting losses on market makers by picking off their quotes, the latter used to call them "SOES bandits," and maintained that the practice forced them to quote larger spreads.

6.4.2 Bid-Ask Spreads and Execution Risk

We first solve for the equilibrium of the limit order market in the simplest case, when the security is riskless ($\sigma = 0$), so that traders are only exposed to non-execution risk. If a trader of type y_i trades q shares at price p , he obtains:

$$U(q, y_i, p) = q(\mu_0 + y_i - p).$$

For instance, suppose the trader sells one share (i.e., $q = -1$). This gives a payoff equal to $p - (\mu_0 + y_i)$, which is positive if and only if $p > \mu_0 + y_i$, i.e., if the sale price is higher than the trader's valuation of the asset. As this valuation is never lower than $\mu_0 - L$, no trader will submit a sell order if the transaction price is below $\mu_0 - L$. By symmetry, no trader submits a buy order at a price above $\mu_0 + L$. Therefore, when $\sigma = 0$, we can restrict our attention to the case in which quotes are in the range $[\mu_0 - L, \mu_0 + L]$. For prices in this range, traders of type $+L$ submit only buy orders and those of type $-L$ submit only sell orders. Hence, in what follows, we study only the optimal buy orders for traders of type $+L$ and sell orders for traders of type $-L$.

Let $\hat{A}$ be the maximum ask price that a trader of type $+L$ arriving at time t is willing to take using a buy market order. The trader's strategy is:

$$\begin{aligned} &\text{submit a buy market order if } A_t \leq \hat{A}_t, \\ &\text{submit a buy limit order at price } B_t^* \text{ if } A_t > \hat{A}_t. \end{aligned}$$

Similarly, let $\hat{B}$ be the minimal bid price that a trader of type $-L$ arriving at time t is willing to take with a sell market order. The strategy is:

$$\begin{aligned} &\text{submit a sell market order if } B_t \geq \hat{B}_t, \\ &\text{submit a sell limit order at price } A_t^* \text{ if } B_t < \hat{B}_t. \end{aligned}$$

Suppose that a (high-valuation) investor submits a limit order to buy at bid price B at time t and recall that the successor trader is a low-valuation type with probability $\tau/2$. Given the above strategies for the successor at time $t+1$, the execution probability P_t of the limit order is:

$$\begin{aligned} P_t(B) &= 0 \text{ if } B < \hat{B}_{t+1}, \\ P_t(B) &= \frac{\tau}{2} \text{ if } B \geq \hat{B}_{t+1}, \end{aligned}$$

Hence, if the trader opts for a buy limit order, the optimal bid price is $B_t^* = \hat{B}_{t+1}$, because a lower bid will not execute, and a higher bid yields no improvement in execution probability. Thus, the highest expected utility from a buy limit order is:

$$E_t \left[U(q_{t+1}, +L, \hat{B}_{t+1}) \right] = \frac{\tau}{2} (\mu_0 + L - \hat{B}_{t+1}).$$

Therefore, the cut-off ask price $\hat{A}_t$ that makes the buyer indifferent between a buy market order and a buy limit order solves:

$$\mu_0 + L - \hat{A}_t = \frac{\tau}{2} (\mu_0 + L - \hat{B}_{t+1}) \quad (6.32)$$

The argument for a low-valuation seller is analogous. The seller's optimal ask price is $A_t^* = \hat{A}_{t+1}$, and the cut-off bid price $\hat{B}_t$ that makes the seller indifferent between a sell market order and a sell limit order solves:

$$\hat{B}_t - (\mu_0 - L) = \frac{\tau}{2} (\hat{A}_{t+1} - (\mu_0 - L)). \quad (6.33)$$

As the parameters (τ, μ_0, L) are time-invariant, traders face exactly the same problem in each period. Hence, it is natural to focus on steady-state equilibria in which traders choose strategies that do not depend on time, that is:

$$\hat{B}_t = \hat{B} \text{ and } \hat{A}_t = \hat{A} \quad \forall t. \quad (6.34)$$

We can now easily compute traders' equilibrium bids, offers and cut-off prices by solving equations (6.32), (6.33), and (6.34). We obtain:

$$A^* = \hat{A} = \mu_0 - L + \frac{4L}{2 + \tau}, \quad (6.35)$$

$$B^* = \hat{B} = \mu_0 + L - \frac{4L}{2 + \tau}. \quad (6.36)$$

As τ increases, execution risk decreases. There are two possible reasons why a limit order may not be taken: either trading is terminated (probability $1 - \tau$), or the next period's trader's private valuation is the same as the current trader's (probability $\tau/2$), so there is a non-execution probability $1 - \tau/2$. Thus, in this model parameter τ measures execution risk, which is a key determinant of the bid-ask spread since:¹³

$$A^* - B^* = 2 \left(\frac{2 - \tau}{2 + \tau} \right) L.$$

The bid-ask spread is decreasing in the parameter τ , and thus increasing in execution risk. Indeed, as τ rises, the cost of not trading immediately upon arrival with a market order becomes smaller because the likelihood of a limit order being filled increases. Thus, in equilibrium, limit orders must be more aggressive if they are to attract counterparties; accordingly ask and bid prices become more aggressive as τ gets larger $\left(\frac{\partial A_t^*}{\partial \tau} \leq 0 \text{ and } \frac{\partial B_t^*}{\partial \tau} \geq 0 \right)$.

This effect of execution risk on traders' order placement strategies can explain the fact that, in real-world markets, bid-ask spreads widen at the end of the trading day. Indeed, as the end of the day nears, the likelihood that limit orders will remain unfilled gets higher. This prospect increases market makers' power, as traders are willing to pay greater price concessions to avoid losing a trading opportunity.

6.4.3 Bid-Ask Spreads and Volatility

Now we turn to the more complex case in which the security is risky, so that $\sigma > 0$. Let $\hat{B}_{t+1}(y_{t+1}, v_{t+1})$ be traders' sell cut-off price at time $t + 1$. Intuitively, $\hat{B}_{t+1}(\cdot)$ increases with v_{t+1} and y_{t+1} , other things equal, since traders should demand a higher price to sell when (i) the asset expected payoff is higher or (ii) their valuation for the asset is higher. But it is not clear a priori whether $\hat{B}_{t+1}(-L, v_t + \sigma)$ is higher or lower than $\hat{B}_{t+1}(+L, v_t - \sigma)$. To fix things, suppose that:

$$\hat{B}_{t+1}(-L, v_t - \sigma) < \hat{B}_{t+1}(-L, v_t + \sigma) < \hat{B}_{t+1}(+L, v_t - \sigma) < \hat{B}_{t+1}(+L, v_t + \sigma). \quad (6.37)$$

Under this conjecture, which will be verified in equilibrium, Figure 6.8 shows the execution probability of a buy limit order at price B submitted at

¹³ The structure of the model is such that the limit order book features either a buy limit order at price B^* or a sell limit order at price A^* but not both orders at once. However, as all buy market orders execute at A^* and all sell orders at B^* , it is natural to define the bid-ask spread as $A^* - B^*$.

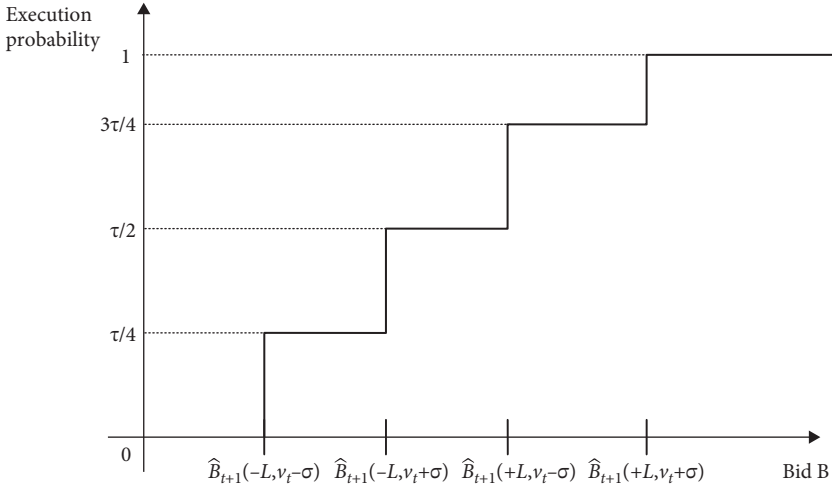


Figure 6.8 Bid price and execution probability.

time t . For example, a buy limit order with a low bid price in the interval $[\hat{B}_{t+1}(-L, v_t - \sigma), \hat{B}_{t+1}(-L, v_t + \sigma)]$ executes if and only if (a) the next trader's private valuation is $-L$ and (b) bad news arises about the asset ($\varepsilon_{t+1} = -\sigma$). This event happens with probability $\frac{\tau}{4}$.

A trader who places a buy limit order loses money if her limit order is hit by traders with a high private valuation ($y = +L$). Indeed, the latter is willing to sell only when her valuation for the asset is below the price of the buy limit order. As this valuation is the highest, this means that the limit order submitter necessarily buys at a price above her own valuation for the asset. Thus, she bears a loss when trading with a trader with a high valuation. This argument rules out any buy limit orders at prices of $\hat{B}_{t+1}(+L, v_t - \sigma)$ or higher, because the resulting increased execution probability is achieved solely by attracting high-private-valuation counterparties in addition to those who would be willing to take a lower bid.

Thus a trader of type $+L$ who places a buy limit order must choose one of only two prices: a conservative low price B_l with a low execution probability of $\frac{\tau}{4}$,

$$B_l \equiv \hat{B}_{t+1}(-L, v_t - \sigma),$$

and a more aggressive price B_h with a greater execution probability $\frac{\tau}{2}$,

$$B_h \equiv \hat{B}_{t+1}(-L, v_t + \sigma).$$

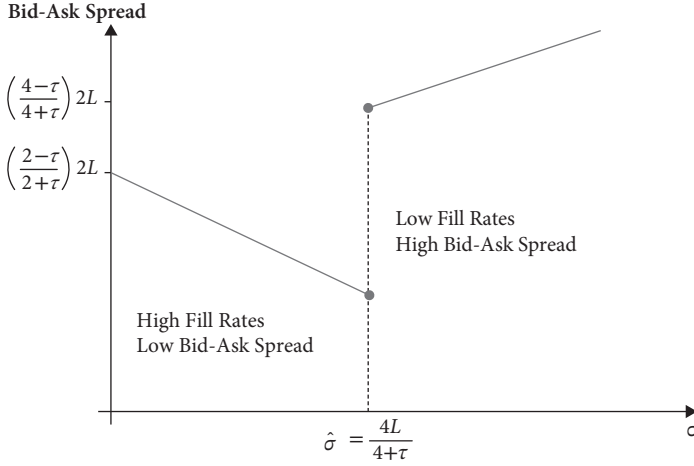


Figure 6.9 Impact of volatility on the bid-ask spread.

Intuitively, the trader faces a trade-off between paying less and having a higher probability of execution. The optimal choice depends on volatility. In equilibrium, traders choose the low-execution-risk strategy if volatility is sufficiently low, so that the utility loss when the asset value drops remains small. Specifically, in equilibrium, traders of type $+L$ adopt the following bidding strategies:

$$\text{post a buy limit order at } B_h = v_t + L - \frac{2(2L - \sigma)}{2 + \tau} \text{ if } \sigma \leq \hat{\sigma},$$

$$\text{post a buy limit order at } B_l = v_t + L - \sigma - \frac{8L}{4 + \tau} \text{ if } \sigma > \hat{\sigma},$$

where the threshold volatility $\hat{\sigma}$ is defined by:

$$\hat{\sigma} = \frac{4L}{4 + \tau}.$$

Similarly, traders of type $-L$ post the following quotes in equilibrium:

$$\text{post a sell limit order at } A_h = v_t - L + \frac{2(2L - \sigma)}{2 + \tau} \text{ if } \sigma \leq \hat{\sigma},$$

$$\text{post a sell limit order at } A_l = v_t - L + \sigma + \frac{8L}{4 + \tau} \text{ if } \sigma > \hat{\sigma}.$$

The method of proof to obtain this result is the same as that used for the case $\sigma = 0$, as explained in the Appendix to this chapter. Of course, if $\sigma = 0$, the expressions for bid and ask prices are identical to those obtained in Section 6.4.2.

The bid-ask spread in equilibrium is:

$$A_h - B_h = 2 \frac{2 - \tau}{2 + \tau} L - \frac{4}{2 + \tau} \sigma \quad \text{if } \sigma \leq \hat{\sigma},$$

$$A_l - B_l = 2 \frac{4 - \tau}{4 + \tau} L + 2\sigma \quad \text{if } \sigma > \hat{\sigma}.$$

Figure 6.9 shows the spread as a function of volatility, σ . Interestingly, in contrast to most other models of bid-ask spreads, the effect of volatility here is non-monotonic. The reason is that an increase in σ raises the cost of being picked off, but also reduces the market power of those who submit limit orders.

To see this, recall that when $\sigma \leq \hat{\sigma}$, investors optimally choose limit orders with a high execution probability, i.e. that execute whether the asset value goes down or up. Now an increase in volatility raises the lowest price (e.g., $\hat{B}_{t+1}(-L, v_t + \sigma)$) required by investors to sell when the value goes up. As a result, if a buyer wants to retain a high execution probability, he must submit limit orders at increasingly high prices as volatility increases. By symmetry, a seller must submit limit orders at increasingly low prices. In other words, an increase in σ reduces the market power of those submitting limit orders. Thus, when $\sigma \leq \hat{\sigma}$ the bid-ask spread declines as volatility increases, and as a result, limit order investors obtain less and less surplus.

At $\sigma = \hat{\sigma}$, this surplus is so small that it becomes more profitable for limit order traders to submit less aggressive quotes at the cost of a lower execution probability. As noted earlier, such quotes only execute when there is bad news. Hence, they are shaded by the amount of the adverse movement that triggers their execution, so when $\sigma > \hat{\sigma}$ the spread is increasing in σ .

When $\sigma \leq \hat{\sigma}$, limit orders have a higher execution probability than when $\sigma > \hat{\sigma}$. There are more market orders relative to limit orders, and the volume of trading is higher. This means that markets where limit orders are relatively more prevalent ($\sigma > \hat{\sigma}$) are not necessarily more liquid. Indeed, such intensive use of limit orders may simply reflect the fact that traders find it too costly to submit a market order, as in the model for $\sigma > \hat{\sigma}$.

6.4.4 Indexed Limit Orders, Monitoring, and Algorithmic Trading

Pegged limit orders. The risk of being picked off makes the market more illiquid. One solution to this problem is to peg limit orders to changes in the fair value of the security. That is, traders' limit orders can be automatically "repriced" when

there is a change in the fair value of the security, and so protected against pick-off risk. The equilibrium is therefore identical to the case in which $\sigma = 0$.¹⁴

In practice, some trading platforms allow indexed limit orders. Since market organizers do not know the fair value of the security, they peg the prices of indexed limit orders to the price of some other, related security (e.g., an index) or to the quotes posted for the same security on other platforms.¹⁵ But the protection offered by this form of indexing is imperfect and does not fully eliminate the risk of limit orders being picked off.

Monitoring limit orders. Traders can also mitigate pick-off risk by monitoring. For instance, if bad news arrives a buyer can quickly cancel his limit order and resubmit it at a lower price, and similarly in case of good news he can resubmit it at a higher price to keep a high execution probability. If this monitoring is costless, then we are back to the case where $\sigma = 0$. In practice, however, monitoring is not costless (it takes work) and trading platforms sometimes charge a fee for order cancellations.

The cost of monitoring limit orders has declined considerably in recent years, as traders increasingly automate their routing decisions to get better protection against pick-off and non-execution risk. Traders also invest to reduce trading latency, the time it takes for their messages (e.g., a request to cancel an order) to be processed by the exchange. For instance, they pay fees to have their servers close to trading platforms (a practice known as colocation). However, automation can also be used to pick off stale limit orders more quickly. Thus, its net effect on liquidity depends on whether it is a greater help to those who seek protection against the risk of being picked off or to those who want to pick off stale quotes. We analyze this evolution in details in Chapter 9.

Box 6.3 Limit Orders as Free Options

There is an interesting analogy between options and limit orders. For instance, submitting a sell limit order is similar to writing an American call option: it enables investors to buy a security at a “strike” price equal to the ask price of the limit order.¹⁶ Market participants want to exercise these options

¹⁴ As Figure 6.9 shows, authorizing pegged limit orders does not necessarily minimize the spread. Indeed, in the model, suppressing limit order traders’ exposure to changes in the asset value can increase their market power, as explained at the end of the previous section.

¹⁵ For instance, limit orders in a given stock traded on the NYSE can be pegged to the best bid and ask prices for this stock on the NYSE via the use of Primary Pegged Orders. Pegged limit orders are then automatically repriced every time there is a change in the best bid or ask price, so that their limit price remains at a fixed distance from the best price.

¹⁶ A buy limit order, by symmetry, is similar to an American put option.

when they are “in the money,” that is, when the price attached to a sell limit order is below the fair value of the security.

The options implicit in limit orders are “free,” since the investors who submit them receive no payment for “writing” (selling) them. Yet, these options can be profitable because, unlike actual options, limit orders sometimes execute even when they are “out the money,” simply because some investors (liquidity traders) need to buy the security (say, to hedge), even in the absence of solid information about its value. Limit order traders must therefore price their orders to balance the expected loss when the implied option goes into the money and is exercised against the expected gain when it is exercised out of the money.

This balancing act is very similar to that of market makers who compensate for the cost of trading with better informed investors with the revenue from dealing with liquidity traders, as seen in Chapter 3 (see also Copeland and Galai 1983). There are some significant differences, though: the exposure to pick-off risk also depends on how fast the trader can access the market and how intensively he monitors his quotes. Thus, the liquidity effect of pick-off risk is linked to investment decisions in routing and monitoring technologies. Moreover, competition between traders seeking to pick off limit orders forces them to race to do so as soon as the option is even slightly in the money, thus endogenously forcing accelerated exercise of the option and limiting its potential value.

6.4.5 Order Flow and the State of the LOB

In the previous sections we have analyzed how execution risk and pick-off risk affect the bid-ask spread, the composition of the order flow, and order aggressiveness. However, the model remains too simple to illuminate the relationship between the depth of the LOB and order flow. In particular, it does not allow us to study how variations in depth affect the aggressiveness of traders' orders. Yet intuitively, the depth of the book on both sides should affect order placement strategies.

Consider a buyer. If a large number of shares are offered at the best bid, then the likelihood of quick execution with a limit order is low. In this case, the buyer is more likely to submit a limit order that improves upon the best bid price, or even a market order. That is, an increase in depth on the buy side of the book makes aggressive buy orders more likely.

Similarly, sellers are more likely to submit a market order when the LOB is deep at the best ask price. Hence, when the number of shares offered at the best

ask price increases, the execution probability of buy limit orders improves, other things equal. That is, an increase in the depth on the sell side of the LOB makes aggressive buy orders less likely.

In this section, we modify the model developed above to formalize these intuitions. Suppose there is no uncertainty about the time at which the asset pays off, and that we count time backward. That is, $t = 1$ denotes the last period of the trading day, $t = 2$ the penultimate period, and so on. For simplicity, consider only the last three periods. The results generalize to an arbitrary number of periods.

To simplify the analysis further, we assume that traders must position their limit orders on a grid with tick size Δ . The first quote on the grid below the fair value is $b = \mu_0 - \frac{\Delta}{2}$, the first above it is $a = \mu_0 + \frac{\Delta}{2}$. The tick size is such that:

$$\frac{\Delta}{2} < L < \Delta. \quad (6.38)$$

Thus, investors of type $+L$ never trade at a price above a and investors of type $-L$ never trade below b . Submitting a limit order with a price outside the range $[b, a]$ is never optimal, since it has zero execution probability. As we shall see, several buy limit orders will queue at price b , several sell limit orders at a . Time priority is used to determine the sequence in which they are filled.

In addition, a market maker stands constantly ready to sell shares at a and buy shares at b . This market maker does not have priority of execution over limit orders placed at the same price. Thus, arriving traders face a simple choice. Consider a trader of type $+L$ for instance. Since he can always trade immediately at price a using a market order, his choice boils down to a market order filled at price a or a limit order to buy at b .

If a trader of type $+L$ submits a buy market order at time t , he obtains:

$$U(1, L, a) = L - \frac{\Delta}{2}.$$

If he submits a buy limit order instead, he obtains:

$$U(1, L, b) = P_{bt}(n_{bt}, n_{at}) \left(L + \frac{\Delta}{2} \right),$$

where $P_{bt}(n_{bt}, n_{at})$ is the execution probability of a buy limit order submitted at time t when the book has n_{bt} buy limit orders (each for one share) at price b and n_{at} sell limit orders (each for one share) at price a (not counting the shares offered by the market maker).

Thus, submitting a buy limit order is optimal if and only if:

$$P_{bt}(n_{bt}, n_{at}) > \frac{2L - \Delta}{2L + \Delta}. \quad (6.39)$$

We obtain the same decision rule for a seller. That is, a trader of type $-L$ optimally submits a sell limit order at price a if and only if:

$$P_{st}(n_{bt}, n_{at}) > \frac{2L - \Delta}{2L + \Delta}, \quad (6.40)$$

where $P_{st}(n_{bt}, n_{at})$ denotes the execution of a sell limit order at price a at time t when the state of the book is $\{n_{bt}, n_{at}\}$.

Hence, the decision for a limit order depends on execution probability, which varies over time, insofar as non-execution gets more likely as the end of the trading day approaches, and the state of the book varies.

Equilibrium. In equilibrium, the execution probability of a limit order at a given time in each state of the LOB depends on future traders' order placement decisions. In turn, these decisions depend on execution probabilities. At time $t = 1$, however, execution probabilities are zero for limit orders, since the trader arriving at this time is the last. Thus, we can solve for traders' optimal actions at each time by "backward induction": first, we derive traders' optimal decision at time $t = 1$; we obtain limit order execution probabilities at time $t = 2$, and using conditions (6.39) and (6.40), we infer traders' optimal orders in each state of the book. We can then proceed to time $t = 3$.

Time $t = 1$. If the trader arriving at time $t = 1$ were to place a limit order, it would certainly not execute, since he is the last trader of the day. Thus, the trader submits a buy market order if he is type $+L$ and a sell market order if he is type $-L$.

Time $t = 2$. Suppose that the trader arriving at $t = 2$ is a buyer ($y_i = +L$). Given the equilibrium actions of the trader arriving at time $t = 1$, the execution probabilities are:

$$\begin{aligned} P_{b2}(0, n_{a2}) &= \frac{1}{2}, \\ P_{b2}(n_{b2}, n_{a2}) &= 0 \text{ if } n_{b2} \geq 1. \end{aligned}$$

Notice that $\frac{2L - \Delta}{2L + \Delta} < \frac{1}{2}$ since $L < \Delta$. Thus, if $n_{b2} = 0$, condition (6.39) is satisfied and the investor optimally submits a buy limit order; otherwise, a buy market order. The optimal strategy of a seller is symmetric: a sell market order if $n_{a2} \geq 1$ and a sell limit order if $n_{a2} = 0$.

Time $t = 3$. Finally, we derive the optimal strategy of the trader arriving at time $t = 3$. Again, suppose that this trader is a buyer. Given traders' optimal actions at times $t = 1$ and $t = 2$, we have:

$$P_{b3}(0, n_{a3}) = \frac{3}{4} \text{ if } n_{a3} \geq 1,$$

$$P_{b3}(0, n_{a3}) = \frac{1}{2} \text{ if } n_{a3} = 0,$$

$$P_{b3}(1, n_{a3}) = \frac{1}{4} \text{ if } n_{a3} \geq 1,$$

$$P_{b3}(1, n_{a3}) = 0 \text{ if } n_{a3} = 0,$$

$$P_{b3}(n_{b3}, n_{a3}) = 0 \text{ if } n_{b3} \geq 2.$$

Thus, the buyer arriving at time $t = 3$ submits a buy limit order at b if $n_{b3} = 0$. If $n_{b3} = 1$ and $n_{a3} \geq 1$, the optimal order placement strategy depends on L/Δ . If $\frac{L}{\Delta} \leq \frac{5}{8}$, then a buy limit order is optimal since its execution probability ($1/4$) is such that condition (6.39) is satisfied. Otherwise, a buy market order is optimal. In other states of the LOB, a trader of type $+L$ submits a buy market order, since a limit order has zero execution probability. The strategy of a seller is symmetric. Table 6.1 summarizes the optimal action for a buyer arriving at time $t = 3$ for each possible state of the book and the execution probability of a buy limit order in each state, when $\frac{L}{\Delta} \leq \frac{5}{8}$. The table shows the optimal order for a buyer arriving at time $t = 3$ ("lo" denoting limit order and "mo" market order) and the execution probabilities of a buy limit order at this time for each possible state of the LOB.

Order flow and state of the book. The equilibrium obtained in this example has several interesting properties (see Table 6.1). First, at each time the execution probability of a new limit order decreases as the number of shares offered on that side increases (e.g., at time 3, $P_{b3}(0, 1) = \frac{3}{4} > P_{b3}(1, 1) = \frac{1}{4}$), and it increases

Table 6.1 Optimal Order Submission Strategy and Execution Probabilities for a Buyer at $t = 3$

n_{a3} :	n_{b3} :	0	1	≥ 2
0		(lo, $\frac{1}{2}$)	(mo, 0)	(mo, 0)
1		(lo, $\frac{3}{4}$)	(lo, $\frac{1}{4}$)	(mo, 0)
2		(lo, $\frac{3}{4}$)	(lo, $\frac{1}{4}$)	(mo, 0)
3		(lo, $\frac{3}{4}$)	(lo, $\frac{1}{4}$)	(mo, 0)
≥ 4		(lo, $\frac{3}{4}$)	(lo, $\frac{1}{4}$)	(mo, 0)

along with the number of shares on the opposite side (e.g., $P_{b3}(0, 1) = \frac{3}{4} > P_{b3}(0, 0) = \frac{1}{2}$). These properties extend to any time. For instance, at all times, $P_{bt}(n_{bt}, n_{at})$ decreases in n_{bt} and increases in n_{at} .

The first property follows from the fact that limit orders queuing at the same price execute in the sequence in which they were submitted (time priority). Hence traders are more likely to submit market orders when the LOB on their side is deeper. But this means that a limit order has a better chance of execution when the book on the other side is deeper, which is the second property. Thus, at any given time, a buy limit order is more likely when the book on the sell side is deep and less likely when the buy side is deep. Several empirical studies (e.g., Biais, Hillion, and Spatt 1995; Griffiths et al. 2000; Ranaldo 2004) confirm this prediction. Using data from the Swiss Stock Exchange, Ranaldo (2004) finds that traders submit more aggressive orders (higher-priced limit orders or market orders) when the LOB is deeper on their side and less aggressive orders when it is deeper on the opposite side. In the same vein, in a pioneering study of the Paris Bourse, an electronic limit order market, Biais, Hillion, and Spatt (1995) observe that: "Improvements in the best quote are especially frequent when the depth at the quotes is already large. This reflects the trade-off between the execution probability and price" (p. 1657).

These properties of execution probability imply that successive orders submitted to the market are not independent, even though traders' types (buyer/seller) are not serially correlated. To see this, let $I_{bt} = 1$ if a buy limit order is submitted at time t , and $I_{bt} = 0$ otherwise. Similarly $I_{st} = 1$ if a sell limit order is submitted at time t , and $I_{st} = 0$ otherwise. In equilibrium, we have:

$$Pr(I_{k2} = 1 | I_{k3} = 1) \leq Pr(I_{k2} = 1 | I_{k3} = 0) \text{ for } k \in \{b, s\}.$$

That is, the likelihood of two consecutive limit orders in the same direction (e.g., buy-buy) is smaller than that of two limit orders in opposite directions (e.g., buy-sell).¹⁷ In other words, the model predicts that the direction of limit orders should be negatively serially correlated.¹⁸ Suppose that a buy limit order arrives at time t . This order increases the depth at the best bid price, making it less attractive for the next trader to also submit a buy limit order and more attractive to submit a sell limit order.

The model also implies a positive serial correlation in the direction of trades (remember that this is defined as the direction of the market order side). The

¹⁷ Here, we just consider dates $t = 3$ and $t = 2$ since the model has only three dates. But this property of the model is more general.

¹⁸ For instance, suppose that $n_{b3} = 0$ and $n_{a3} = 0$. If a sell limit order is submitted at time 3, the likelihood of observing a buy limit order at time $t = 2$ is $\frac{1}{2}$. In contrast, if a buy limit order is submitted at time 3, the likelihood of observing a buy limit order at time 2 is zero.

intuition is as follows. Suppose that a buy market order arrives at time t . The depth of the book on the ask side decreases by one share. As a consequence, the execution probability for an investor submitting a sell limit order improves. Sell limit orders become more attractive and the likelihood that the next investor will submit a sell market order consequently declines. Hence, the likelihood of observing a buy market order at time $t - 1$ is greater given a buy market order at time t than given a sell market order. For instance, suppose that one share is offered on each side of the book at time 3 ($n_{b3} = n_{a3} = 1$) and a sell market order now arrives. Then the likelihood of another sell market order at time 2 is $\frac{1}{2}$ in equilibrium. If instead a buy market order hits the market at time 3 then the likelihood of a sell market order at time 2 is nil. Thus, the likelihood of two consecutive transactions in the same direction is greater than that of two transactions in opposite directions.

These properties imply that variations in the state of the LOB can generate time-series dependence in the order flow endogenously, even though traders' types (buyer/seller) are not correlated over time. In fact, in limit order markets orders are serially correlated. For instance, the likelihood of an order of a given type at a given time (say, a buy market order) is greater following the arrival of an order of the same type—which is partly consistent with the model presented here, where two consecutive transactions are more likely to be in the same direction than in opposite directions. But the model presented here does not explain why, empirically, a buy (sell) limit order is more likely after another buy (sell) limit order. This effect may stem from the fact that traders often split large orders into small consecutive trades, a strategy that is outside the model considered in this section.

6.5 Further Reading

Garman (1976) and Cohen et al. (1981) provide early formal analyses of dynamic limit order markets. But they specify traders' beliefs on the execution probabilities of their limit orders exogenously.

Glosten (1994) develops the first equilibrium model of trading in a limit order market. His model is similar to that developed in Section 6.2.3 but he assumes that a pro-rata allocation rule is used to determine how limit orders tied at the same price are executed (as in Section 6.3.2). He also assumes that the number of traders submitting limit orders is infinite. Biais, Martimort, and Rochet (2000) relax this assumption and characterize the equilibrium of the limit order market when there is imperfect competition among limit order traders (see also Baruch and Glosten (2019)).

Using data from the Stockholm Stock Exchange, Sandas (2001) tests the specification of the model considered in example 1 (in its more elaborate version with asymmetric information). He tests whether snapshots of LOBs collected at the time of each transaction satisfy the restrictions imposed on cumulative depth and quotes by equations (6.16) and (6.17). He rejects the model, principally because the LOB predicted by the model appears too deep relative to the actual LOBs that he observes. One possible explanation is that competition among limit order traders is imperfect, so that the zero profit conditions that characterize the competitive equilibrium are not satisfied in reality.

In the model considered in Section 6.2.3, informed investors are assumed to submit market orders. Bloomfield, O'Hara, and Saar (2005) run laboratory experiments in which informed and uninformed participants trade in a limit order market. They find that informed traders sometimes use limit orders. Brogaard et al. (2019) find that limit orders significantly contributes to price discovery, which also suggests that informed investors sometimes use limit orders. Kaniel and Liu (2006) and Rosu (2020) develop theoretical models in which informed traders submit limit orders.

Seppi (1997) introduces the model of limit order trading without asymmetric information considered in Section 6.2. He extends this model to account for adverse selection and analyzes the last-mover advantage of the NYSE specialist (see Section 6.3.3). Parlour and Seppi (2003) use this model to study competition between a hybrid and a pure limit order market. In this framework, Foucault and Menkveld (2008) analyze competition between two pure limit order markets (see Section 7.4.2).

Seppi (1997), Viswanathan and Wang (2002), and Back and Baruch (2007) compare limit order markets and uniform price auctions. Back and Baruch (2007) show that a dynamic uniform auction in which informed investors can split their market orders over time always has an equilibrium that is identical to that obtained in Glosten (1994).

Dynamic models of limit order trading have been developed by Foucault (1995); Parlour (1998); Foucault (1999); Goettler, Parlour, and Rajan (2005); Foucault, Kadan, and Kandel (2005); Van Achter (2006); Rosu (2009); Large (2009); Hoffmann (2014); and Dugast (2018). The model presented in Section 6.4.1 is based on Foucault (1999). Foucault, Kadan, and Kandel (2005) and Rosu (2009) study the role of impatience and time to execution. Parlour (1998) offers a systematic equilibrium treatment of the relationships between the depth of the LOB and the aggressiveness of order submission, as discussed in Section 6.4.5. Buti and Rindi (2013) and Buti, Rindi and Werner (2017) build on this model to study theoretically the role of hidden orders and dark pools in limit order book markets. Goettler, Parlour, and Rajan (2005) develop a more general version of the framework considered in Section 6.4.1, allowing traders to submit

multiple limit orders when they arrive, and to stay for more than one period, and drawing traders' private valuations from a continuous distribution. They develop an algorithm to find a stationary Markov-perfect equilibrium of the limit order market numerically and run it to derive various comparative statics. In particular, they consider the effect of reducing tick size. They show that a smaller tick reduces limit order traders' surpluses but increases investors' total surplus. Goettler et al. (2009) apply the same methodology to analyze the decision to acquire information in a limit order market.

A natural question is whether order choices that are observed in reality can be explained by the trade-offs described in Section 6.4.1. Hollifield, Miller, and Sandas (2004) test this, using a more general version of that framework. Their test consists in estimating execution probabilities and pick-off risks for limit orders. Then they check whether observed limit orders maximize expected payoffs.¹⁹ They do not reject the model when they test it for buy and sell orders separately.

Hollifield et al. (2006) use this framework to measure the efficiency of a limit order market empirically (the Vancouver Stock Exchange). They find that the limit order market is quite efficient, in that about 90% of the maximum gain from trades is realized, according to their estimates. Interestingly, unexecuted limit orders constitute the main source of inefficiency.

There is also a rich empirical literature on the determinants of order submission choices (see, for instance, Griffiths et al. 2000; Ellul et al. 2007; Rinaldo 2004) and the references therein) and the cost and benefits of using market and limit orders (see Handa and Schwartz 1996; Harris and Hasbrouck 1996). Parlour and Seppi (2008) provide a survey of the literature on limit order markets.

6.6 Appendix. Volatility and Make or Take Decisions: Derivations

Here we briefly sketch the derivation of the equilibrium of the model of Section 6.4.3, which extends the model presented in Section 6.4.2 when $\sigma > 0$. As for $\sigma = 0$, we consider steady state equilibria in which investors' strategies do not depend on time. Assume first that $\sigma \leq \hat{\sigma}$. In this case, in equilibrium, buyers and sellers post only limit orders with high execution probability, that is: $B^* = \hat{B}(-L, v_t + \sigma)$ and $A^* = A_t = \hat{A}(+L, v_t - \sigma)$. Thus, traders' cutoff prices are:

$$\mu_0 + L - \hat{A}(+L, v_t) = \frac{\tau}{2} (\mu_0 + L - \hat{B}(-L, v_t + \sigma)), \quad (6.41)$$

$$\hat{B}(-L, v_t) - (\mu_0 - L) = \frac{\tau}{2} (\hat{A}(+L, v_t - \sigma) - (\mu_0 - L)). \quad (6.42)$$

¹⁹ They show that this can be checked by testing a simple monotonicity condition, using the estimates obtained for execution probabilities and pick-off risk.

Further, it must be that the expected payoff with aggressive (high-execution-probability) orders is greater than that obtained with conservative (low execution probability) limit orders. Thus the following inequalities must hold:

$$\frac{\tau}{2} (\mu_0 + L - \hat{B}(-L, v_t + \sigma)) \geq \frac{\tau}{4} (\mu_0 + L - \hat{B}(-L, v_t - \sigma)) \quad (6.43)$$

$$\frac{\tau}{2} (\hat{A}(+L, v_t - \sigma) - (\mu_0 - L)) \geq \frac{\tau}{4} (\hat{A}(+L, v_t + \sigma) - (\mu_0 - L)) \quad (6.44)$$

Investors' equilibrium cutoff prices are obtained by solving the system of equations (6.41) and (6.42). This yields closed-form solutions for B_h and A_h . Armed with these solutions for the cut-off prices, we can verify that conditions (6.37), (6.43), (6.44) are satisfied if and only if $\sigma \leq \hat{\sigma}$. The derivation of the equilibrium when $\sigma > \hat{\sigma}$ follows similar steps; see Foucault (1999) for a full analysis.

6.7 Exercises

1. Deriving a competitive LOB. Consider the model developed in section 6.2.3. We make the following parametric assumptions:

1. The trader who arrives in period 1 knows the final value of the security v with probability π . Otherwise, he is uninformed.
2. If the trader who arrives in period 1 is uninformed, he buys or sells (with equal probability) a number of shares x that has an exponential distribution with parameter θ . That is, the size distribution of the market order submitted by an uninformed trader arriving in period 2 is $f(x) = \frac{1}{2} \theta e^{-\theta|x|}$.
3. The final value of the security in period 2 has the following probability distribution: $g(v) = \frac{1}{2\sigma} \exp\left(-\frac{|v-\mu|}{\sigma}\right)$. This implies that σ is the mean absolute deviation of v and $E(v|v \geq z) = z + \sigma$.
4. The tick size is nil ($\Delta = 0$).

Let $Y(A)$ be the cumulative depth up to ask price A in the book and A^* be the lowest ask price in the LOB.

- a. Show that when $v \geq A^*$, the optimal strategy of the informed trader is to buy $Y(v)$ shares.
- b. Using this observation and the zero profit condition (6.13), show that in equilibrium:

$$Y(A) = \frac{1}{\theta} \left[\ln\left(\frac{1-\pi}{\pi}\right) + \ln\left(\frac{A-\mu}{\sigma}\right) + \frac{A-\mu}{\sigma} \right] \text{ if } A > A^*.$$

- c. Show that the book becomes thinner on the ask side when (i) π increases or (ii) σ increases. What is the economic intuition for this result?

2. **Time priority vs. random tie-breaking rule.** Consider example 2 in Section 6.2.3 but assume that the tick size Δ is strictly positive, such that:

$$A_1 = \mu + \Delta < \mu + \sigma < \mu + 2\Delta.$$

Time priority is enforced as in the baseline model of Section 6.2.3.

- a. Explain why the LOB will feature at least q_L shares offered at price A_2 .
- b. Let Y_1 be the number of shares offered at price A_1 . Define $r = \sigma/\Delta$. Observe that $r \in [1, 2]$. Using the assumptions regarding the order flow at time 1, show that:

$$(a) Y_1 = 0 \text{ iff } \frac{(r-1)\pi}{1-\pi} \geq 1.$$

$$(b) Y_1 = q_s \text{ iff } \frac{(r-1)\pi}{1-\pi} \in [1-\phi, 1).$$

$$(c) Y_1 = q_L \text{ iff } \frac{(r-1)\pi}{1-\pi} \in [0, 1-\phi).$$

- c. Why does Y_1 decrease with π ?
- d. Now assume that $\frac{(r-1)\pi}{1-\pi} \in [1-\phi, 1)$ and suppose that time priority is not enforced any more. Instead, if two traders post a limit order at price A_1 , then the offer that is executed first is determined randomly. Specifically, the limit order posted by trader $j \in \{1, 2\}$ is executed first with probability 0.5. Let Y_1^j be the number of shares offered by trader $j \in \{1, 2\}$ at price A_1 . Explain why, in equilibrium, Y_1^1 and Y_1^2 must satisfy the following conditions for $j = 1$ and $j = 2$:

$$\begin{aligned} & (A_1 - E(v|q \geq Y_1^j)) \Pr(q \geq Y_1^j) \\ & + (A_1 - E(v|q \geq Y_1^1 + Y_1^2)) \Pr(q \geq (Y_1^1 + Y_1^2)) \leq 0, \end{aligned}$$

with a strict inequality if $Y_1^1 = Y_1^2 = 0$.

- e. Suppose that $q_L = 2q_s$. Deduce that $Y_1^1 + Y_1^2 = q_s$ form an equilibrium if:

$$\frac{(r-1)\pi}{1-\pi} \in \left[(1-\phi), \frac{1+(1-\phi)}{2} \right],$$

when the “random” tie-breaking rule is used.

- f. Why is cumulative depth greater when the random tie-breaking rule is used for $\frac{(r-1)\pi}{1-\pi} \in \left[(1-\phi), \frac{1+(1-\phi)}{2} \right]$?
3. **Time priority vs. pro-rata allocation.** Consider the model developed in Section 6.2.2 and suppose $C < A_1 - v_0$. The size of the incoming market order (in absolute value) has a uniform distribution on $[0, Q]$, that is,

$$F(q) = \frac{q}{Q}.$$

- a. Show that in this case the cumulative depth at price A_k is:

$$Y_k = Q \left(1 - \frac{C}{A_k - \mu} \right), \forall k.$$

- b. Now suppose that instead of time priority, a pro-rata allocation rule is used, as described in Section 6.3.2. Further assume that $A_1 - \mu > 2C$. Then show that the cumulative depth at price A_1 is $Y_1^r = \frac{(A_1 - \mu)Q}{2C}$.
- c. Why does the pro-rata allocation rule yield greater cumulative depth at all ask prices?
4. **Competition among specialists and liquidity.** Consider the model of Section 6.3.3 and assume that two specialists can stop out a market order. When a market order arrives, they post a stop-out price at which they are ready to fill. The specialist with the more competitive price executes the order. If there is a tie, the order is split equally between the two specialists.
- a. Show that if:

$$\frac{q_L}{q_S} > 1 + \frac{\pi}{(1 - \pi)(1 - \phi)},$$

then, in equilibrium, the offers in the LOB are as described by equations (6.25) and (6.26), and the specialists stop out the small orders at a price (bid or ask) equal to μ .

- b. In this case, do the specialists improve liquidity?
5. **Make/take fees and bid-ask spreads.** Consider the model of section 6.4.1 with $\sigma = 0$ and $\tau = 1$. As Chapter 7 explains, trading platforms often charge different fees for market and limit orders. Let f_{mo} be the fee per share paid by a market order placer and f_{lo} the fee per share paid by a limit order placer when the limit order executes (there is no entry fee for limit orders). Finally let f be the total fee earned by the platform on each trade, that is, $f = f_{mo} + f_{lo}$.
- a. Compute bid and ask quotes in equilibrium.
- b. Show that the bid-ask spread decreases in f_{mo} and increases in f_{lo} . Explain.
- c. Trading platforms often subsidize traders who submit limit orders. That is, they set $f_{lo} < 0$ and $f_{mo} > 0$, maintaining that this practice ultimately helps to narrow the spread and benefits traders submitting market orders. Holding the total trading fee f fixed, is this argument correct?

6. Cryptocurrency limit order books. The data for this exercise are contained in the Excel file *Ch6_ex6_data.xlsx*, available on the companion website for the book. The Excel file has three worksheets: *BTCUSD_Quotes* contains a snapshot of the limit order books for Bitcoin-USD (BTCUSD) on July 1, 2019, *ETHUSD_Quotes* contains a snapshot of the limit order book for Ethereum-USD (ETHUSD) on March 1, 2020 on the Coinbase exchange, and *BTCUSD_Trades* provides the first 10,000 trades in BTCUSD on July 1, 2019. To facilitate comparisons of the limit order books for Bitcoin and Ethereum, all amounts of cryptocurrencies in *BTCUSD_Quotes* and *ETHUSD_Quotes* are expressed in dollars, that is, number of units of the relevant currency multiplied by the price in each case (in the original data, these amounts were in units of each cryptocurrency). In contrast, in *BTCUSD_Trades*, amounts are in units of Bitcoin.

- a. Produce a table with summary statistics (the mean, median, standard deviation, and number of observations) for the trade amount and price (using data in the sheet *BTCUSD_Trades*) and the bid and ask price for BTCUSD (using data in the sheet *BTCUSD_Quotes*).
- b. For both BTCUSD and ETHUSD, calculate the cumulative dollar amounts that can be purchased or sold at the various bid and ask prices posted in the limit order book of each currency and represent them graphically in a single figure, placing prices on the vertical axis and quantities (measured in dollars) on the horizontal axis, as in Figure 6.5. The one-period LOB model with asymmetric information covered in example 1 of Section 6.2.3 predicts that there should be a strictly positive bid-ask spread even for very small order sizes. Is this true in both markets?
- c. The model in Section 6.2.3 predicts that the bid-ask spread should be equal to $\frac{2\lambda}{\theta}$. If we assume that θ is the same in both markets, based on the relative bid-ask spread found in your answer to question **b**, which of the two markets has more informed trading?
- d. Another prediction of Section 6.2.3 is that market depth should be inversely related to the orders' informativeness. According to your answer to question **c**, which market should have the deeper order book? Estimate the bid and ask market depth for both markets by calculating the total amount of liquidity available 5% away from the midprice. For example, to calculate the depth on the bid side of BTCUSD, find the total dollar amount of Bitcoin that could be sold up to price \$10222 ($= 0.95 \times 10760.7$). Are the estimates of market depth in line with the theoretical prediction?

- e. At different points in Chapter 6, we assumed that the bid and ask side are symmetric. Using your answer to question d, does this assumption seem to hold in the data? What could be an explanation for why it might not hold?
- f. We have often assumed that the market order is exponentially distributed. Using the trade data in BTCUSD, plot the empirical distribution together with an exponential distribution with parameter of $\lambda = 6.13$.²⁰ Is this specification of the exponential distribution a good fit for the empirical distribution? (Note that outliers have been removed from the trade data so that no trade amounts are higher than 2.)

²⁰ This parameter is chosen so that the mean of the exponential distribution, $1/\lambda$, matches the mean of the sample distribution.